



Tracking the Evolution of the Loss and Damage Financing Architecture under the UNFCCC

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DECLARATION

I, K R Hari Krishna, hereby declare that this dissertation entitled 'Tracking the Evolution of the Loss and Damage Financing Architecture under the UNFCCC' as the outcome of my own study undertaken under the guidance of Dr Kamal Kumar Murari, Centre for Climate Change and Sustainability Studies, School of Habitat Studies, Tata Institute of Social Sciences, Mumbai. It has not previously formed the basis for the award of any degree, diploma, or certificate of this Institute or of any other institute or university. I have duly acknowledged all the sources used by me in the preparation of this dissertation.

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CERTIFICATE

This is to certify that the dissertation entitled “**Tracking the Evolution of the Loss and Damage Financing Architecture under the UNFCCC**” is the record of the original work done by **Hari Krishna K R** under my guidance and supervision. The results of the research presented in this dissertation/thesis have not previously formed the basis for the award of any degree, diploma, or certificate of this Institute or any other institute or university.

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LIST OF ABBREVIATIONS

Abbreviation	Expanded Form
AIIB	Asian Infrastructure Investment Bank
AOSIS	Alliance of Small Island States
AR6	Sixth Assessment Report (IPCC)
ARC	African Risk Capacity
BCR	Benefit-to-Cost Ratio
BIM	Barbados Implementation Modalities
C3S	Copernicus Climate Change Service
CBA	Cost-Benefit Analysis
CBDR-RC	Common but Differentiated Responsibilities and Respective Capabilities
CCRIF	Caribbean Catastrophe Risk Insurance Facility
CDA	Critical Discourse Analysis
CMA	Conference of the Parties serving as the Meeting of the Parties to the Paris Agreement
CO ₂	Carbon Dioxide
COP	Conference of the Parties
CP	Conference of the Parties (in decision numbering, e.g., 2/CP.19)
CPI	Climate Policy Initiative
ELD	Economic Loss and Damage
EM-DAT	Emergency Events Database
EMDE	Emerging Market and Developing Economies
EU	European Union
EUR	Euro (currency)
FRLD	Fund for Responding to Loss and Damage
G77	Group of 77 (developing countries coalition)
GBP	British Pound Sterling
GDP	Gross Domestic Product
GEF	Global Environment Facility
GHG	Greenhouse Gas
GNP	Gross National Product
IBRD	International Bank for Reconstruction and Development
INC	Intergovernmental Negotiating Committee
IPCC	Intergovernmental Panel on Climate Change
L&D	Loss and Damage
LDC	Least Developed Country
LDCs	Least Developed Countries
MDB	Multilateral Development Bank
MDBs	Multilateral Development Banks

NDCs	Nationally Determined Contributions
NELD	Non-Economic Loss and Damage
NPV	Net Present Value
OECD	Organisation for Economic Co-operation and Development
SBI	Subsidiary Body for Implementation
SBSTA	Subsidiary Body for Scientific and Technological Advice
SCCF	Special Climate Change Fund
SIDS	Small Island Developing States
UAE	United Arab Emirates
UK	United Kingdom
UN	United Nations
UNDP	United Nations Development Programme
UNEP	United Nations Environment Programme
UNFCCC	United Nations Framework Convention on Climate Change
US	United States
USD	United States Dollar
WIM	Warsaw International Mechanism
WMO	World Meteorological Organization

ABSTRACT

Loss and damage (L&D) was central to climate talks for more than two decades before being formally set up in the Warsaw International Mechanism (2013) and Article 8 of the Paris Agreement (2015). However, neither of these milestones led to a clear definition of L&D, nor an understanding of what it involves or who is accountable for it. This paper argues that this lack of clarity is deliberate and that it stems from a continuous argument between developed and developing nations where language was the main tool of political strategy.

Using Fairclough's three-part model for critical discourse analysis, this thesis examines Decision 2/CP.19 and how it alters the power dynamic between developed and developing nations under the UNFCCC. The analysis operates on three levels: the specific word choices within the text itself, the negotiating positions and history of the institutions that created it, and the wider set of discourses it belongs to.

Developing nations, especially AOSIS, used the demand for compensation as a tool to make rich countries formally recognise L&D as a separate issue from adaptation. Developed countries countered this by focusing on disaster risk reduction and questioning the science of climate change attribution to avoid liability. The agreed-upon language in 2/CP.19 is deliberately vague, representing a pause rather than an agreement. While the push for direct compensation lessened after the Paris Agreement, the fundamental call for climate justice remains.

Keywords: loss and damage; climate change negotiations; critical discourse analysis; Warsaw International Mechanism; compensation; climate justice; UNFCCC

INTRODUCTION

Background

Global average temperatures have breached critical thresholds, actively destroying economic assets and human lives (Germanwatch e.V., 2025). With Earth already recording its hottest day on 22 July 2024, and with 2024 recording the highest average temperature on record, 1.5°C above pre-industrial levels, there is a necessity to either curb or adapt to the increase in temperature (C3S, 2025; WMO, 2024).

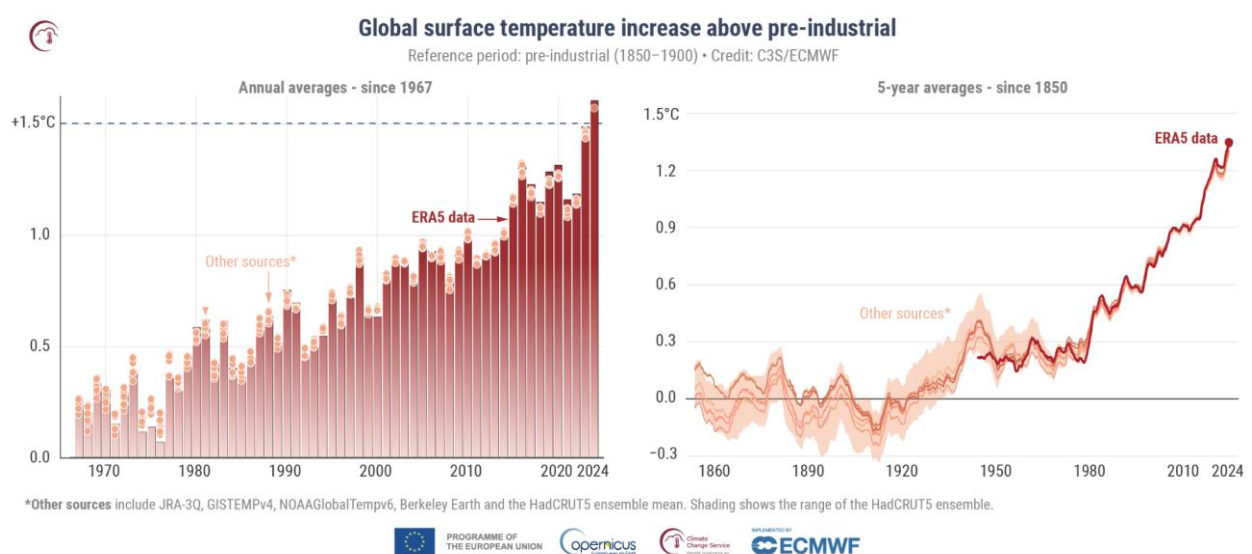


Figure 1.1: Global surface air temperature increases above the 1850–1900 pre-industrial reference period, shown as annual averages since 1967 (left) and as 5-year averages since 1850 (right) (C3S, 2025).

Anthropogenic greenhouse gas emissions drive this warming, actively altering the intensity, frequency, and duration of destructive climatic events, and, under current policy pathways, global temperatures will rise by 2.6°C to 3.1°C by 2100 (UNFCCC, 2025b). These warming triggers severe impacts through extreme weather events and slow-onset processes. While extreme weather events involve sudden, discrete events such as tropical cyclones, flash floods, and heatwaves, and destroy physical infrastructure, kill populations, and disrupt supply chains instantly, slow-onset processes evolve gradually over years or decades without a clear start or end point (D. Coumou et al., 2024; Fischer & Knutti, 2015; Limsakul et al., 2024). However, rather than operating in isolation, these mechanisms compound and exacerbate one another (Limsakul et al., 2024). Rising sea levels, for example, raise the baseline water level, allowing

storm surges from tropical cyclones to penetrate farther inland and cause disproportionate coastal flooding (Hegglin et al., 2022; Thomas et al., 2020).

The economic costs of these impacts are severe, unresolved by current adaptation efforts, and accelerating. Researchers project that residual climate-related economic losses and damage in developing countries will range from \$290 billion to \$580 billion annually by 2030, and by 2050, these annual costs will range from \$1.132 trillion to \$1.741 trillion (Bhandari et al., 2022; Markandya & González-Eguino, 2019). The global economy already suffers from this acceleration, as climate-related disasters inflicted \$320 billion in direct economic losses in 2024 (MunichRE, 2025). While these estimates remain conservative because they measure only direct physical damage, they often fail to capture indirect economic losses such as reduced labour productivity, supply chain failures, and unmonetised ecosystem destruction.

Historical data from EM-DAT, compiled by Delforge and team (2025), confirm the devastating cumulative toll of these climate hazards. Between 01 January 2000 and 14 April 2026, the world experienced over 9,803 distinct extreme weather events, resulting in over 15,71,197 people killed globally, with direct, inflation-adjusted economic losses from these events reaching nearly \$4.8 trillion. Storms caused the greatest economic destruction, accounting for 49.6% of total losses at \$2.4 trillion, followed by floods, which caused 21% of total losses at \$1.1 trillion.

These aggregate figures obscure extreme local vulnerabilities, as single high-impact events frequently obliterate the entire economic output of exposed nations. In 2017, Hurricane Maria struck Dominica, causing up to \$1.8 billion in damage, equivalent to 270% of the island nation's annual Gross Domestic Product (GDP) (Germanwatch e.V., 2025). Similarly, Cyclone Pam devastated Vanuatu in 2015, inflicting economic damages equal to 64% of its national GDP (Germanwatch e.V., 2025). While SIDS and LDCs are more susceptible to such hazards, larger developing economies also suffer massive proportional losses. In 2022, extreme monsoon rainfall caused flooding across one-third of Pakistan, affecting 33 million people, killing over 1,700, and causing nearly \$15 billion in direct damage, destroying roughly 10% of the nation's GDP (Germanwatch e.V., 2025). These extreme financial shocks force vulnerable countries to take on risky foreign debt to secure immediate disaster relief, thereby stripping their fiscal space and crippling long-term economic growth.

Three Pillars of Climate Action and Finance

Mitigation and Adaptation

The international community has responded to this deterioration through Mitigation and Adaptation in climate action and finance. **Mitigation targets the physical causes of climate change and requires reducing greenhouse gas (GHG) emissions and enhancing carbon sinks** (UNEP, 2023c). Governments implement mitigation through renewable energy transitions, energy efficiency improvements, and carbon markets to reduce emissions and lower excess GHG levels in the atmosphere.

While the Paris Agreement aims to limit global warming to 1.5°C above pre-industrial levels, global efforts consistently fall short, as evidenced by the current Nationally Determined Contributions (NDCs), which project a temperature increase of 2.6°C to 3.1°C by 2100 as mitigation fails to stabilise the climate (UNFCCC, 2015a). While efforts to reduce greenhouse gas emissions and adapt to climate impacts have made some progress, there is growing recognition that these measures alone cannot fully prevent or mitigate the damage caused by climate change, particularly in areas that need adaptation.

Adaptation involves structural and behavioural adjustments to reduce human and ecological vulnerability to expected climate impacts, including communities building sea walls to hold back rising tides, planting drought-resistant crops to secure food supplies, and installing early warning systems to evacuate populations before cyclones make landfall (UNEP, 2023b). While mitigation prevents and reduces emissions, adaptation projects, such as sea walls or irrigation systems, protect existing economic assets from future damage, thus preserving value and avoiding future costs, which economists can calculate as a financial benefit (UNEP, 2023a).

Loss and Damage

Loss and Damage (L&D) refers to the actual and/or potential manifestation of adverse effects due to climate change that people cannot cope with or adapt to (Huber & Murray, 2024; McDonnell, 2023; Nand et al., 2024; UNFCCC Secretariat, 2012; Warner & van der Geest, 2013). L&D covers impacts from sudden, extreme weather events and gradual, slow-onset events, particularly in developing countries, which are more vulnerable to such impacts (UNFCCC, 2014). In this context, loss consists of effects that cannot be fixed easily, leading to irreversible changes, while damage refers to harmful impacts where restoration is possible

but leads to high costs and burdens for the affected (Bhowmik et al., 2021; Doktycz & Abkowitz, 2019; Nand et al., 2024).

The idea of "Loss and Damage" is, in essence, a product of the limitations of the conventional pillars of climate action, i.e., mitigation and adaptation. L&D refers to the adverse effects of climate change, which are often inevitable and irreversible, and persist even if, or even in the absence of, successful implementation of the above two strategies (UNDP, 2023; UNEP, 2023d).

L&D can thus be established as a product of "residual risk" with impacts documented as oftentimes irreversible and include economic and non-economic damages such as the loss of homes and livelihoods due to erosion, the degradation of agricultural land, trauma resulting from extreme weather events, and the desecration of sacred sites due to sea-level rise (UNDP, 2023). These residual impacts are essential, however, as knowledge of the limits to adaptation, i.e., the effort required to adapt to 4°C rather than a lesser degree of warming, can inform the amount and pace of global mitigation required globally (Klein et al., 2014).

Types of L&D

Categorisation based on Valuation Methodology

Loss and Damage are generally categorised into two main types: economic and non-economic forms (Bahinipati, 2020; Nand et al., 2024; UNFCCC Secretariat, 2012).

Economic Loss and Damage (ELD) impacts resources, goods, and services. This does include tangible losses such as damage to crops, public and private property, reduced income, and loss of farm employment (Bahinipati, 2020; García-Portela, 2024). Other economic impacts include land turning to desert, loss of livestock, additional irrigation costs, and damage to existing infrastructure and other assets (Bahinipati, 2020; Boda et al., 2021).

Non-Economic Loss and Damage (NELD) covers impacts on life that do not have a direct market value or are not traded in markets (typically being public goods or more abstract items), with some examples being the loss of cultural heritage, traditional knowledge, and biodiversity (Bahinipati, 2020; Fezzigna et al., 2025; García-Portela, 2024; Nand et al., 2024; van Schie et al., 2024). Human-related NELD consists mainly of intangible and indirectly tangible costs such as loss of life, physical and mental health, mobility, conflicts over water, declining groundwater levels, and harm to forests or water quality (Bahinipati, 2020; Gewirtzman et al., 2018; Nand et al., 2024; van Schie et al., 2024). It can also include changes in productive land,

emotional distress, the sacrifice of celebrations, temporary relocation, and permanent changes in one's profession (Bhowmik et al., 2021).

NELD is highly context-specific and often difficult to define, measure, or price (Boda et al., 2021; Nand et al., 2024; Scown et al., 2025; van Schie et al., 2024). NELD is also frequently overlooked and inadequately addressed in policies, making it a significant blind spot in many current assessments, as it is difficult to conceptualise, account for, and monetise (Bahinipati, 2020; Doktycz & Abkowitz, 2019; Huber & Murray, 2024).

Developed nations generally report higher absolute economic L&D (L&D) due to their extensive capital, infrastructure, and insured assets (Bahinipati, 2020; Boda et al., 2021; Boyd et al., 2021; Fezzigna et al., 2025). However, developing countries often face a greater economic L&D when measured as a percentage of their national income or GDP because their economies are less able to absorb economic shocks, and their livelihoods often depend heavily on natural capital (Bahinipati, 2020; Dorkenoo et al., 2022; Fezzigna et al., 2025; Martyr-Koller et al., 2021). For instance, a study conducted by Bahinipati (2020) in rural India found that farmers experienced high immediate tangible and intangible losses, highlighting the disproportionate impact of climate-related disasters on more vulnerable economies and populations, where even minor absolute losses can represent a significant portion of their limited resources.

Categorisation based on Prevention and Risk Reduction

Loss and Damage can also be understood by considering whether effective climate action could have lessened their effects, to grasp the importance of taking meaningful steps to combat climate change and its consequences.

Firstly, **Avoidable effects** refer to the negative impacts that could be minimised or prevented through planned adaptation options through substantial mitigation and adaptation efforts (Bahinipati, 2020; Mechler et al., 2020).

Meanwhile, **Unavoided effects** result from missed opportunities for planned adaptation actions, but are constrained by technical, financial, or socio-economic constraints (Bahinipati, 2020; Bhowmik et al., 2021; Mechler & Deubelli, 2021; Warner & van der Geest, 2013).

Finally, **Unavoidable effects** refer to the impacts of climate change that cannot be stopped, even after trying to reduce or adapt to it and are often called "residual impacts", these occur

when societies hit their limits to adaptation (Aleksandrova & Costella, 2021; Bahinipati, 2020; Bhowmik et al., 2021, 2021; Fekete & Sakdapolrak, 2014).

L&D and Adaptation Limits

The Intergovernmental Panel on Climate Change (IPCC) explains that these limits occur when a society can no longer achieve its objectives, protect its values, or avoid risks deemed intolerable, even with adaptation measures (Calvin et al., 2023). There are two main types of adaptation limits:

1. **Hard limits** mean no adaptation options are physically possible to avoid intolerable risks, and when crossed, any action will not be effective, leading to permanent and irreversible harm (Calvin et al., 2023; Mechler & Deubelli, 2021; Roberts & Pelling, 2018).
2. **Soft limits** occur when adaptation options are unavailable due to a lack of funding, administrative or technical capacity, or other socio-economic barriers. However, specific development interventions can sometimes postpone or remove these soft limits (Calvin et al., 2023; Mechler et al., 2020; Mechler & Deubelli, 2021; Roberts & Pelling, 2018).

L&D goes beyond adaptation because it involves situations in which action becomes ineffective, leads to maladaptation through adverse outcomes, or is simply not possible due to concerns beyond adaptation's scope (Robinson et al., 2021; van Schie et al., 2024). Even if efforts are successful, some residual L&D is expected, and these often lead to permanent losses.

Importance of L&D

L&D has gained increasing attention and is recognised as a distinct area alongside mitigation and adaptation in international climate policy frameworks like the Warsaw International Mechanism (WIM) and Article 8 of the Paris Agreement, through the motto "avert, minimise, and address" being a core idea of Loss and Damage (UNFCCC, 2014, 2016b).

Addressing L&D is thus very important because climate change impacts are becoming more frequent and severe worldwide (Doktycz & Abkowitz, 2019; Gewirtzman et al., 2018; Huber & Murray, 2024). Communities already vulnerable, especially in developing countries, are affected the most, even though they have contributed the least to global warming, and many of these climate impacts cannot be reversed and go beyond what current efforts can handle (Boyd

et al., 2021; Calvin et al., 2023; Gewirtzman et al., 2018; Huber & Murray, 2024; McDonnell, 2023; Nand et al., 2024; van Schie et al., 2024). If not measured correctly and accounted for, the actual costs of disasters and a community's ability to recover are severely underestimated, showing the need for specific financial support and reliable systems to help affected populations (Bahinipati, 2020; Doktycz & Abkowitz, 2019).

Currently, there is too much focus on economic costs, which overlooks the significant non-economic impacts on people's well-being and social structures (Bahinipati et al., 2024; Doktycz & Abkowitz, 2019; Gewirtzman et al., 2018; McDonnell, 2023). Understanding L&D thus helps us to identify gaps in data, biases in research, and the urgent need for more representative studies in vulnerable regions and by giving a complete picture of risks, policymakers can create better plans for reducing risks and building community resilience (Bahinipati et al., 2024; Clarke et al., 2023).

The Financing Problem of L&D

The global climate finance architecture cannot manage permanent, unpriceable destruction. Multilateral development banks (MDBs), bilateral donors, and private investors supply the vast majority of international climate capital, and they strictly operate on a market-return logic, favouring projects that generate revenue or protect existing capital (CPI, 2023). Consequently, mitigation projects and adaptation infrastructure dominate the funding landscape, leaving little explicitly earmarked for L&D (CPI, 2023; UNFCCC, 2025b).

Mitigation attracts hundreds of billions of dollars annually because it creates profitable assets. In 2021 and 2022, global mitigation finance reached USD 1.15 trillion, of which the majority was funded by private capital through project-level market-rate debt and equity (CPI, 2023), producing a clear, calculable Return on Investment.

Adaptation finance operates on a different, but equally market-driven, logic by preserving the value of the targeted resource. These interventions aim to shield existing economic assets from projected climate hazards (CPI, 2023).

Economists, such as Mechler and team in 2019, justify these investments using Cost-Benefit Analysis (CBA), calculating Net Present Value (NPV) and Benefit-to-Cost Ratios (BCR). For example, constructing a ring dike around a coastal city requires immense capital, but it prevents future flood damage to real estate and infrastructure. The avoided future cost functions as a tangible financial benefit. Because adaptation protects future economic output, development

banks and governments can justify the upfront costs. When institutions do fund adaptation, they rely heavily on debt. Between 2016 and 2022, loans constituted the majority of climate financing provided by MDBs (CPI, 2023). Forcing highly vulnerable nations to borrow to rebuild after unavoidable disasters worsens their existing debts, strips them of fiscal power, and creates a vicious cycle of ruin.

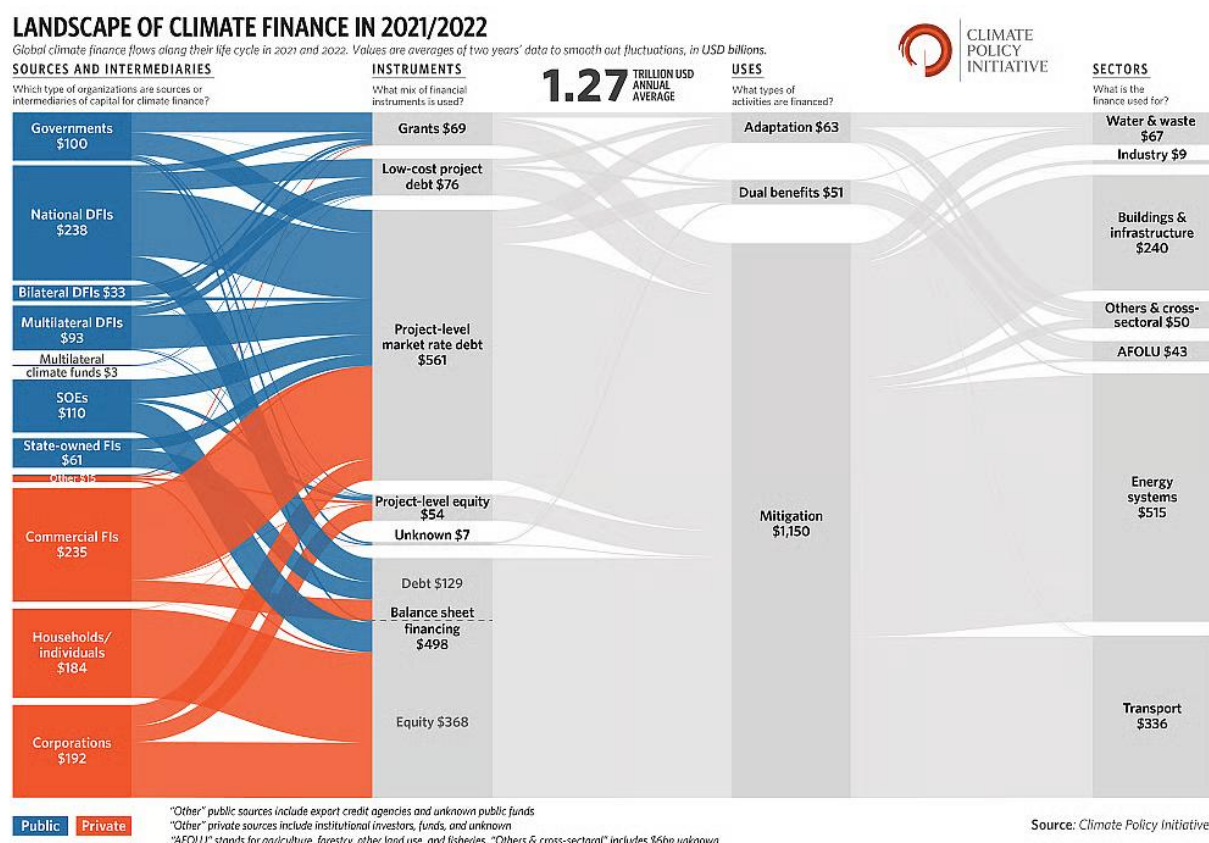


Figure 1.2: CPI in 2023 showed that global climate finance has doubled to \$1.27 trillion annually but remains insufficient (1% of global GDP) and unevenly distributed, with developing nations receiving minimal funding despite urgent needs.

L&D finance, on the other hand, fundamentally violates both the revenue-generation logic of mitigation and the value-preservation logic of adaptation. As rebuilding a hospital destroyed by a Category 5 hurricane does not generate new economic growth, such interventions simply restore assets rather than expanding capital or yielding financial return. Consequently, private investors and commercial banks refuse to fund L&D recovery.

Since L&D is the realisation of residual risks, this can create a unique fiscal challenge as it generates no revenue, and conventional financial instruments do not work when applied to it. Equity investments require an underlying asset that appreciates or generates dividends. A

submerged coastal village or a salinised agricultural plain, for example, offers no equity to hold, leaving loans as the primary conventional tool.

However, deploying loans for L&D creates a severe structural injustice. Currently, multilateral development banks and bilateral donors provide 63% of all adaptation-specific finance in the form of debt, according to the UNEP Adaptation Gap report in 2023. Applying debt to L&D forces highly vulnerable nations to borrow money to recover from disasters they did not cause.

For residual risks, the system relies on insurance mechanisms such as catastrophe bonds and risk-pooling facilities. These include catastrophe bonds and sovereign risk-pooling facilities like the African Risk Capacity (ARC) and the Caribbean Catastrophe Risk Insurance Facility (CCRIF) (Adeniyi, 2025; Gewirtzman et al., 2018; UNDP, 2023). Catastrophe bonds transfer the financial risk of natural disasters to capital markets, offering investors high yields while providing governments with quick liquidity if a specific disaster triggers a payout.

However, this reliance on commercial insurance fails to account for the physical realities of L&D. While insurance is effective for rare, sudden events such as storms, it is not effective for slow-onset events such as sea-level rise or desertification, because they are sure to occur. Insurance cannot be used for events that are guaranteed to occur (Gewirtzman et al., 2018; Robinson et al., 2021; Verchick, 2018). In 2015, severe floods struck Malawi, causing US \$366 million in economic losses. Nevertheless, the disaster triggered an insurance payout of only US\$ 8.1 million, highlighting that even when vulnerable states secure coverage against hazards, payouts consistently fall short of actual losses (Adeniyi, 2025).

When extreme weather destroys a nation's infrastructure, its Gross Domestic Product (GDP) falls instantly. To fund immediate emergency relief and rebuild essential services, the government must issue new sovereign debt, often at punitive, market-rate interest, which strips the nation of its fiscal space. The country would therefore be required to divert domestic tax revenues away from education, healthcare, and future climate adaptation to service high-interest foreign debt. Over time, recurring climate disasters trap the nation in a cycle of insolvency, permanently halting its long-term economic development through a sovereign debt crisis. Conventional debt instruments thus do not resolve L&D; they merely convert physical and infrastructural losses into prolonged financial liabilities and debts.

The financing problem intensifies when addressing NELD. Since it encompasses the intangible and abstract losses, it defies conventional economic models because these losses are strictly

incommensurable (Mechler et al., 2019). Financial markets require assets to possess a uniform metric of exchange, typically currency, to facilitate trading and valuation. However, no financial instrument can rebuild a lost cultural identity, replace an extinct language, or compensate a community for the permanent submergence of its sovereign territory (Allen et al., 2024; Mechler et al., 2019). The value of these elements stems from essential human and environmental relationships, making traditional Cost-Benefit Analyses inadequate, as they require monetary valuation of irreplaceable entities.

Attempting to force NELD into conventional financial instruments also raises profound ethical concerns, as assigning a monetary value to human life or culture distorts the true scale of the losses and damages involved. If an economic model prices human life or health based on lost future wages, it implies that a life lost in a high-income nation is worth more than one lost in a Least Developed Country. Additionally, such financial models ignore NELD because it lacks a market price, underestimating the scale of residual climate risk. Consequently, current market-based mechanisms treat these profound human and ecological losses as invisible externalities, rendering them entirely unfindable through conventional funding methods.

The operational mechanics of mitigation, adaptation, and market returns prove that L&D requires a categorically different financial architecture. Because L&D interventions address permanent losses, yield zero financial return, and involve incommensurable non-economic losses, they must be funded through curative, debt-free grants.

A new financial model is therefore needed for L&D, one that is primarily based on grants, which do not need to be paid back necessarily.

The Research Problem

This particular financial need drives the political discourse on responsibility, but different parties have taken different positions over time. Developing countries insist on a liability system that mandates funding. Developed countries, on the other hand, prefer to offer voluntary support grounded in "cooperation" and "solidarity," ensuring that L&D finance does not entail legal liability or a right to compensation. The problem of L&D finance is thus how to secure adequate, reliable funding for irreversible harm without the legal admission of guilt that developed nations refuse to accept (Calliari et al., 2018; Yvonne Karimi-Schmidt, 2020).

This tension is evident in the Paris Agreement as shown by **Decision 1/CP.21, which explicitly states to “not involve or provide a basis for any liability or compensation” (UNFCCC,**

2015b, 2016a). With voluntary financial support, funding cycles are unpredictable and depend on fiscal conditions and the political will of donor nations, rather than being tied to historical responsibility or the scale of the damage inflicted on the vulnerable (Åberg & Jeffs, 2022; van Schie et al., 2024). With no justification for levies or taxes being imposed on historical emitters, this further pushes L&D and Climate Finance in general towards traditional grant-based aid (Pill, 2022; Wewerinke-Singh & Salili, 2020).

The operationalisation of the FRLD at COP28 mirrors this shift from obligation to cooperation with the fund, hosted by the World Bank rather than being operated as an independent entity with assessment powers, along with donor preferences for voluntary contributions and stringent fiduciary oversight, rather than the other disbursement methods favoured by developing nations (Schalatek, 2025).

The research problem thus centres on the current L&D financing architecture, which was intended to provide substantial, predictable support to address the irreversible consequences of climate change. However, the framework does not allow for liability and compensation, thereby preventing a direct connection between responsibility and financial support. This makes the mobilisation of revenues dependent on voluntary rather than mandatory means. The institutional question, therefore, revolves around the system's ability to generate the necessary revenue.

This study thus seeks to establish whether the shift from a liability-based to a cooperation-based system has affected the authority of the L&D financing framework to mobilise revenues, particularly the FRLD's capacity to mobilise funding.

Aim

The study aims to track the development of the concept of L&D within the UNFCCC and assess the extent to which the shift from 'liability and compensation' to 'cooperation and facilitation' structurally limits the Fund for Loss and Damage (FRLD) 's potential to mobilise revenue.

Objectives

1. Building a timeline of primary sources of L&D policy documents, decisions, draft negotiating texts, and country submissions from the initial 1991 AOSIS proposal to the 2024 operationalisation of the FRLD,
2. Analysing primary sources of L&D in the UNFCCC from two periods: pre-Paris (1991-2015) and post-Paris (2016-2024),

3. Creating a map of country-level negotiating positions to assess which countries shifted from mandating compensation to embracing solidarity contributions,
4. Creating a map of the range of financial instruments proposed in the UNFCCC negotiations vs. those included in the final form of the Fund for Loss and Damage, and the amount pledged in each case.
5. Evaluate notable clauses in the FRLD Governing Instrument to determine the legal limits to the revenue authority's power on mandatory contributions.

Research Questions

Does the shift in the UNFCCC texts from a focus on liability to one on cooperation structurally limit the FRLD's authority to enforce mandatory revenue measures?

Sub-questions

1. How does the use of words like "compensation," "restitution," and "insurance pool" change in the negotiation texts across the time periods?
2. Which country groups altered their positions on financial obligation in the shift from the Warsaw International Mechanism (WIM) to the FRLD?
3. Which financial measures were eliminated in the final decisions based on liability concerns?
4. Does the FRLD Governing Instrument provide the FRLD with legal authority to enforce mandatory capitalisation targets or levies?
5. Which legal and financial limitations on "innovative sources," such as taxation, can be found in the legal *modus operandi* of the FRLD?

Scope

The scope of the study is the institutional and textual evolution of L&D finance under the UNFCCC. The study will cover various official decision texts, draft negotiating texts, country submissions, and governing documents from 1991 to the present. The study will also cover various proposed and adopted financial instruments, as well as publicly disclosed pledges and contributions.

The study will specifically focus on the content analysis of the definition of financial authority as enunciated under the law. The study will also focus on the evolution of terms and changes in the framing of authority and revenue across the different negotiation periods.

Relevance

The study is relevant because it tries to catalyse predictable, sufficient funding to address actual and permanent damage that has already been done. The L&D costs expected in the developing world are estimated at between 290 and 580 billion dollars annually by 2030. However, with current pledges of only 700-800 million dollars, there is a need to better understand the current legal architecture to mobilise resources and cover the funding gap or the permanent solvency problems that arise from the deletion of clauses (Adil et al., 2025; Markandya & González-Eguino, 2019). If the treaty text constrains revenue-raising authority, the problem is structural rather than short-term.

From an academic viewpoint, this study also examines how treaty text influences institutional authority and associated financial outcomes. The study also contributes to institutional economics by examining whether a voluntary contribution system can overcome free-rider problems. Lastly, it contributes to public finance by examining whether a grant-based climate fund, without revenue-raising authority, can be financially viable.

Boundaries

The study will not evaluate the efficacy of the various projects funded under the L&D finance mechanisms. The study will also not cover damage or economic loss estimation or attribution studies. Instead, the study will focus specifically on the institutional level and will cover only textual analysis. The study will focus on the textual evolution of L&D finance under the UNFCCC.

LITERATURE REVIEW

Normative Frameworks in L&D

Liability Frameworks

As established, the failure of current climate finance mechanisms stems from treating climate disaster relief as humanitarian aid, effectively transferring the physical and financial consequences of historical high-emission economic growth from industrialised nations to the Global South. This is the fundamental injustice that the liability model, which requires the causes of damage to pay for the repair, is designed to address.

The liability model rests on the philosophical foundation of corrective justice. Corrective justice, as defined by Wallimann-Helmer et al. (2018), establishes a direct moral and legal relationship between the actor who causes harm and the victim. When an identifiable entity inflicts an injury through wrongful conduct, that entity incurs a strict duty to make the victim whole, with the perpetrator required to restore the victim to the **status quo ante**, i.e., the exact position they would have occupied had the harm never occurred. Applied to climate change, this logic dictates that the historical emissions of industrialised nations caused the climate shifts now devastating developing countries. Therefore, these heavy emitters bear a strict obligation to fund the resulting L&D. In 1991 (UNFCCC, 1991b, 1991a), and later in 2008, through **Decision 1/CP.13** (UNFCCC, 2008), the Alliance of Small Island States (AOSIS) asserted that compensation for climate impacts is a legal entitlement based on historical responsibility, proposing mechanisms in which mandatory contributions from industrialised nations would fund compensation for sea-level rise and extreme weather, directly linking payments to emissions levels and economic capability.

International law codifies this through the doctrine of state responsibility and the **no-harm rule**, requiring states to prevent activities within their jurisdictions from causing significant transboundary harm (Mechler et al., 2019; Simlinger & Mayer, 2018). This principal traces back to the *Trail Smelter Case (United States v. Canada)* in 1941, which established that no state has the right to use its territory to cause severe injury to another state. If a state breaches this obligation by emitting excessive greenhouse gases, it commits an internationally wrongful act. Under the standards established by the *Factory at Chorzów Case* in 1928, full reparation

requires the state to wipe out all consequences of the illegal act, typically via financial compensation when physical restitution is impossible. Consequently, climate-vulnerable states argue that the failure of industrialised nations to curb emissions breaches the no-harm rule, triggering a legal duty to compensate for the resulting L&D.

Domestic environmental frameworks popularised this mechanism as the **polluter pays** principle, which advocates internalising the social and environmental costs of their actions rather than passing them on to the public (Ohdedar, 2016). In the international arena, this principle exerts immense political force because it assigns clear moral culpability to high-emitting nations (Kumar, 2024; Siegele, 2023). Theorists argue that it faces limitations when applied to historical emissions, as early emitters were often unaware of the harm they caused and are now dead (García-Portela, 2023). To resolve this, some scholars propose the **Beneficiary Pays Principle**, which shifts the focus from the act of polluting to the economic benefits it generates (Wallimann-Helmer et al., 2018). Despite this theoretical refinement, the polluter pays principle remains the dominant political narrative because it directly aligns with the corrective justice demand to hold wrongdoers accountable.

However, enforcing strict liability for global climate damage faces a significant practical barrier, as corrective justice requires a direct causal link between a specific wrongful act and a specific injury. Climate change operates through cumulative emissions over decades, making it complex to attribute a single extreme weather event to a single nation's emissions (Mechler et al., 2019). However, since extreme event attribution relies on probabilistic models rather than direct, linear causation, strict liability models fail under conditions of diffuse, cumulative causation (Mechler et al., 2019). Industrialised nations exploit this gap to contest their obligation to pay for specific damages, keeping the liability model politically powerful while leaving it legally unresolved.

Cooperative Model

To resolve this political deadlock over liabilities, international climate frameworks increasingly rely on the **cooperation-or-solidarity model**, grounded in **distributive justice**. Where corrective justice looks backwards to identify who caused harm and assigns liability, distributive justice looks forward and asks how the community should allocate burdens fairly to correct an unfair baseline distribution of resources and risks. Under this model, severe climate impacts become undeserved harms due to geographic bad luck, rather than the direct result of wrongful emissions by specific states. Consequently, financial support becomes a

humanitarian response to severe misfortune rather than a legal obligation to repair an inflicted injury.

This shift in philosophical framing alters the entire basis of financial responsibility. Wallimann-Helmer et al. (2018) show that when L&D is viewed as an undeserved harm requiring redistribution, historical emissions and legal culpability focus on the **ability-to-pay** principle, under which parties with the most resources and excess capacity must contribute the most to the common endeavour, regardless of their historical carbon footprint. This relies on voluntary, progressive contributions driven by current wealth and capacity while abandoning mandatory, assessed contributions tied to a nation's specific emissions record. It thus shifts from rectifying a past wrong to evening out a current unfair disadvantage. As a result, assistance often focuses on the recipient's urgent needs rather than the funder's fault.

Verchick (2018) and Mechler et al. (2019) trace that separating financial support from causation removes the legal mechanism that binds specific polluters to specific damages, moving away from accountability and towards charity. The funding mechanism degrades from restitution, triggered by a legal responsibility, to voluntary aid, dependent entirely on donor willingness and current fiscal capacity. Because donors contribute out of solidarity rather than legal obligation, they retain the power to dictate the terms, timing, and scale of their financial support.

Pill (2022) notes that solidarity offers a highly practical compromise in global negotiations because it recognises the impacts felt by developing countries without assigning fault, thereby avoiding the legal ramifications and endless compensation claims that stall negotiations. Schinko et al. (2018) add that a distributive justice perspective is necessary in the short- to medium-term because attributing specific losses to specific anthropogenic emissions remains scientifically and legally complex. A solidarity fund appeals directly to human morale and expresses compassion, sidestepping the need for international courts or admissions of guilt. However, treating L&D finance purely as humanitarian aid risks double-counting official development assistance and allows developed nations to shirk their historical responsibility for the crisis.

Wallimann-Helmer et al. (2018) defend the distributive justice approach on the grounds of non-ideal theory. Because proving direct causal attribution for every climate disaster remains complex, and enforcing liability remains politically deadlocked, demanding strict corrective justice leaves many victims with nothing. Verchick (2018), drawing on Judith Shklar's work,

argues that injustice flows not only from harmful acts but also from failures to act; "unjustified passivity" by privileged bystanders in the face of disaster is a moral failure in itself. Therefore, wealthier nations owe a duty to act simply because they have the means to prevent or relieve suffering, regardless of the origin of the injury. A distributive approach ensures that communities facing climate threats receive assistance based on their urgent needs. Furthermore, a solidarity framework captures all L&D, providing support regardless of whether a specific disaster stems directly from anthropogenic emissions or natural climate variability.

Wallimann-Helmer (2023) argues that L&D governance under a forward-looking model requires providing vulnerable communities with a sufficient threshold of capabilities to achieve climate resilience. If communities lack the financial and technological resources to recover from extreme weather events, they are entitled to support. This requires comprehensive risk management, early warning systems, and substantial capacity building. Consequently, ensuring a fair baseline distribution of resilience capabilities may require financial transfers as large as those demanded under strict liability.

Ambiguities and Coexisting Frameworks in Treaty Text

Multilateral climate treaties require consensus, yet parties may hold mutually exclusive positions on decisions. To break these deadlocks, diplomats rely on **constructive ambiguity**. As Vanhala (2023) observes, negotiators use deliberately vague language that allows parties with fundamentally incompatible goals to interpret the text in accordance with their own interests.

The treaties intentionally leave the phrase "addressing L&D" undefined. Developing nations read "addressing" as the first step towards a formal liability and compensation framework. Developed nations read the same word as a mandate for technical risk management, insurance schemes, and voluntary aid.

Calliari (2018) notes that vague treaty language systematically favours the party with less to concede. A liability model requires affirmative legal commitments, while the cooperation model requires only the absence of such binding text. Therefore, when negotiators use ambiguous language to bypass a definitive choice, the institutional architecture automatically defaults to voluntary cooperation.

By refusing to assign responsibility, the text strips the framework of enforcement power. The deliberate omission of liability language constitutes a choice that shields developed nations

from mandatory financial contributions. The ambiguity that allows treaties to exist also prevents them from raising capital, laying the foundation for **destructive ambiguity**.

The Political Economy of Climate Finance

Incompatibility with L&D

As Boda et al. (2021) establish, the core definition of L&D, addressing assets, resources, and livelihoods that are irreversibly destroyed, is completely antithetical to the logic of private capital. Private investors require a yield-generating asset or collateral to secure their investments, but neither an inundated coastal village nor a salinised agricultural plain provides either (AIIB et al., 2025; Fritsch et al., 2025). Because L&D interventions simply restore basic survival conditions rather than expanding capital, they cannot generate the revenue streams that commercial banks and private investors demand.

NELD makes private financing even more unworkable as markets lack any pricing mechanism to value lost cultural heritage, indigenous knowledge, or human lives. Because financial instruments demand a uniform metric of exchange, they cannot process or fund the recovery of incommensurable losses. Attempting to force NELD into market-based frameworks distorts the true scale of the harm and leaves the most profound climate impacts entirely unfunded.

This becomes even more obvious when contrasted with the other pillars of climate action. While the mitigation and adaptation sectors successfully attract private capital due to their alignment with market incentives, L&D addresses the residual risks that materialise when adaptation fails, yielding a zero financial return.

Consequently, private capital will not bridge the L&D funding gap. The OECD (2021) confirms that private finance mobilisation for climate action remains highly uneven, flowing primarily to middle-income countries with investment-grade ratings. Private actors consistently refuse to finance projects that deliver public goods without commercial returns.

Concessional Finance and L&D

Multilateral development banks channel the majority of their climate finance through concessional loans (CPI, 2023). However, as Ike Uri et al. (2024) observe, using debt to address L&D forces vulnerable countries to underwrite the cost of climate disasters they did not cause. If a cyclone destroys a small island state's agricultural sector, demanding that the state borrow money to rebuild imposes a direct financial penalty on the victim. Adeniyi (2025) confirms that this "debt-on-disaster" approach is both unjust and destructive, as it increases debt-to-GDP

ratios in countries already facing severe fiscal stress, diverting scarce public funds away from healthcare, education, and future climate adaptation simply to service foreign debt.

Policymakers frequently promote risk transfer mechanisms, such as catastrophe insurance, as the primary alternative to debt, only for debt to fail again. Insurance protects against low-probability, high-impact events that are random and unforeseen. Robinson et al. (2021) and Verchick (2018) demonstrate that this logic collapses when applied to slow-onset events such as sea-level rise, desertification, and the salinisation of freshwater. If a loss is certain, the required premium equals the payout, rendering the insurance mechanism entirely useless for long-term climate impacts.

Furthermore, insurance requires quantifiable damage assessments or specific parametric triggers to release payouts. Gewirtzman et al. (2018) establish that neither method works for NELD. Financial markets lack the mechanisms to price the permanent loss of a cultural heritage site, the erasure of indigenous knowledge, or the destruction of a traditional fishing ground. Because insurance demands a strict monetary metric for exchange, it inherently excludes the most profound human and ecological losses from its frameworks.

Free-Rider Problem in Voluntary Contribution Systems

International climate finance relies heavily on voluntary contributions, creating a severe collective action problem. Without binding legal obligations, the global institutional architecture cannot compel any individual state to contribute capital proportionally to its historical emissions or economic capacity. Consequently, every potential donor has an incentive to withhold funding and wait for other nations to bear the cost, thereby allowing targets to fail.

Åberg and Jeffs (2022) document that major contributors frequently make high-profile pledges to institutions like the Green Climate Fund but deliver only partial disbursements over time. Van Schie et al. (2024) show that these delivery gaps are structurally inherent to voluntary financing models. They do not merely result from political bad faith. When governments face tight budgets, they often ignore international commitments, as there are no penalties for backing out.

If the FRLD operates voluntarily, the free-rider problem is not a risk for administrators to manage, which explains exactly why the normative shift from liability to cooperation matters fiscally. By adopting a voluntary model, the UNFCCC institutionalises delivery gaps,

ensuring that the architecture will systematically underfund the permanent climate losses it was created to resolve.

Documentation Bounding Institutional Authority

International organisations have no power of their own. As legal experts show, a group's authority, what it can do, and its firm limits come only from the multilateral agreements that created it to carry out the specific work its signatories explicitly grant it. If a founding treaty provides that financial contributions are voluntary, the resulting body cannot legally require mandatory funding, compel countries to share costs, or punish states that refuse to pay. In the same way, if the treaty text clearly rules out liability, the body cannot create financial tools based on assessing fault or demanding repayment. The law's text sets the financial facts.

This legal boundary defines what the Fund for Responding to Loss and Damage (FRLD) can do. The FRLD manages the money side of the United Nations Framework Convention on Climate Change (UNFCCC) and the Paris Agreement. The Decision 1/CP.21 explicitly states to “not involve or provide a basis for any liability or compensation” (UNFCCC, 2016). Some, like Gsottbauer et al. (2018) and Wallimann-Helmer et al. (2018), argue that it blocks all compensation claims under the UNFCCC, rendering it unable to deliver climate justice. Others suggest it only limits Article 8 and does not eliminate general rights under international law or future mechanisms for allocating responsibility. Despite these interpretations, the practical outcome within the UNFCCC has been a shift from the “polluter pays principle” to a “victim pays” or “charity” approach.

This statement, however, has more than merely declaratory value and removes the juridical link between causation and compensation, disconnecting from the law on state responsibility. By excluding liability, the UNFCCC framework removed the legal mechanism for vulnerable nations to demand compensation for climate change-related losses, rendering financial support a discretionary political act rather than a binding legal duty (Simlinger & Mayer, 2018).

Because the foundational text removes liability, the FRLD cannot legally calculate specific damages owed by high-emitting states, nor can it compel those states to pay. Furthermore, the COP28 decision establishing the FRLD confirms that contributions remain entirely voluntary.

Literature Gaps Identified

Lack of Long-Term Analysis

The current discussion about L&D focuses on analysing the language used in negotiations, highlighting the divide between rich and poor countries, and examining how the idea of "constructive ambiguity" has changed over time. However, no analysis of all texts on L&D from 1991 to 2024 has been conducted; most research so far has focused only on IPCC reports, not on negotiation texts, leading to a lack of concrete evidence showing how terms related to liability have decreased while terms about cooperation have increased over the last 30 years of talks.

Missing Record of Financial Instruments

Research on financing L&D has concentrated mainly on potential future financial instruments, such as taxes on air travel or bunker fuels. While some historical accounts mention the initial proposal for a 1991 insurance pool, there has not been a comprehensive overview of the financial instruments proposed and then rejected throughout the negotiation process. The current literature suggests avenues for financing L&D, but fails to outline what was actually turned down and the reasons behind these rejections.

Limited Fiscal Authority

There has been insufficient exploration of how the exclusion of "liability" affects the Fund for Responding to Loss and Damage's (FRLD) fiscal authority in collecting contributions or raising funds. These areas have typically been analysed separately rather than as interconnected issues.

Contributions

1. Show the shift from liability-based to cooperation-based approaches to L&D.
2. Identify specific financial instruments that were excluded to highlight changes in financial ambition.
3. Establish the connection between the exclusion of liability and the FRLD's limited fiscal authority.

METHODOLOGY

Research Philosophy

This study cannot use a positivist framework. Positivism assumes an objective reality independent of the observer, one that can be measured, tested, and verified. The FRLD's fiscal authority is a legal arrangement produced entirely by negotiated text, with its meaning and power determined by the words.

This study adopts an **interpretivist epistemology** grounded in a **constructivist ontology**. Constructivism holds that social actors create institutional reality through interaction, negotiation, and text production, rather than discovering a pre-existing one. In the UNFCCC context, "L&D" lacks a fixed natural meaning, as stated in decisions. When developing nations propose "compensation" and developed nations require "assistance," these are not stylistic preferences but are bids to construct an institution with fundamentally different legal powers.

A purely quantitative approach, counting pledges and calculating the funding gap, can describe the outcome but cannot explain it. Explaining why the FRLD was designed without mandatory revenue authority requires reading the texts that authorised it, the texts that preceded it, and the negotiating record that transformed the former into the latter.

Research Design

The central puzzle this paper addresses is how a concept as contested as 'loss and damage' acquired institutional form under the UNFCCC despite the lack of an agreed definition. Answering that question requires a method sensitive to language as a site of political contest, not simply a medium for communicating pre-existing positions. The writing of the texts (decisions, submissions, statements) produced by the UNFCCC negotiation process did not neutrally record political outcomes; rather, it was through the writing itself that the contentious issues surrounding the definition, scope, and responsibility for loss and damage were resolved. This demonstrates how a focus on language as a mere vehicle for pre-existing interests overlooks the very process of interest formation and dispute.

Critical Discourse Analysis is well-suited to this context because UNFCCC negotiations are structurally unequal, with small island states and least-developed countries facing industrialised blocs possessing vastly greater economic weight, legal capacity, and institutional

access. Nevertheless, the outcomes, including the Warsaw International Mechanism and Article 8 of the Paris Agreement, do not straightforwardly reflect this power asymmetry.

Understanding how the weaker parties shaped the outcome necessitates close attention to the relationship between language and power, which is the core concern of critical discourse studies.

Among all CDA methodologies, this paper builds on Fairclough's (1992, 2003) framework rather than, say, the discourse-historical approach of Wodak or the socio-cognitive model of van Dijk. The two methods answer different questions and do not overlap. The CDA component helps answer whether parties frame L&D as a legal wrong requiring a remedy or as a shared misfortune requiring cooperation. What does the shift from "shall compensate" to "should assist" signal about institutional intent? How does removing a word from an operative clause strip an institution of a power it would otherwise hold?

The keyword discourse component systematically shows, across the full corpus, that liability-connected terms declined in discourse over time while cooperation-connected terms rose, thereby strengthening the causal argument. This design is **explanatory** and **descriptive** as it seeks to compare pledges, trace a historical narrative and survey actors.

Theoretical Framework

The theoretical framework draws on Fairclough's (1995, 2003) three-level model of CDA. At the textual level, the study examines specific lexical and grammatical choices, including the substitution of binding verbs (shall, must) with discretionary language (should, encourages), and the replacement of liability-connoting nouns (compensation, restitution) with solidarity-framing terms (assistance, voluntary support).

At the discursive level, the analysis examines how these texts were produced and consumed within the negotiation process, focusing on the specific junctures at which developing-nation proposals for strong legal obligation were diluted to secure consensus. At the social practice level, the study assesses the real-world institutional consequences of these textual shifts, in particular, whether the systematic removal of liability language has legally foreclosed the FRLD's authority to enforce mandatory financial contributions.

Data Sources

The dataset includes four different types of documents created within the UNFCCC framework from 1991 to 2024.

1. The Final adopted COP and CMA Decisions are the official records of the agreements reached, particularly those related to the Warsaw International Mechanism (WIM) and the Fund Mechanism for Loss and Damage (FRLD).
2. Country and Coalition Submissions to the Subsidiary Bodies (SBI/SBSTA) and the Transitional Committee reflect the positions of key groups, including AOSIS, the LDC Group, the African Group of Negotiators, the European Union, and the Umbrella Group.
3. Meeting and Operational Documents of the Fund for responding to Loss and Damage.

The documents were chosen based on two main criteria. First, they must include the phrase “L&D,” which refers explicitly to the UNFCCC policy rather than just harm. Second, texts without this phrase were considered only if they discussed financial tools to address irreversible climate impacts, using terms such as “compensation,” “liability,” “insurance pools,” “risk transfer,” or “solidarity.” The documents that only mentioned “adaptation finance” without recognising the inevitable losses that cannot be adapted to were excluded.

Unit of Analysis	What is Coded	Purpose
Textual Segment	Paragraph or clause containing target keywords; verb modality (shall / should); operative vs preamble context	Qualitative discourse assessment
Financial Pledge	Individual commitment in USD. Pledged/committed/disbursed. source type (sovereign/philanthropic/innovative)	Quantitative financial analysis; delivery gap identification
Negotiating Position	Country or bloc submission; coded liability-supporting, cooperative-only, or neutral/ambiguous	Political analysis; tracking of demand migration

Table 3.1: Details on the unit of analysis and their purpose

Periodisation

The corpus divides at 2015 with the pre-Paris period (1991-2015) covering the emergence of the L&D concept under the UNFCCC and Kyoto Protocol, culminating in the Paris Agreement. The post-Paris period (2016-2024) covers operationalisation with Article 8 work programmes, the Santiago Network, the Glasgow Dialogue, the Transitional Committee process, and the establishment of the FRLD. The 2015 dividing line is not arbitrary, as Decision 1/CP.21 formally excluded liability and compensation from the FRLD's legal mandate. This will serve as a point of reference for analysis.

Analytical Framework

This analysis uses a five-part causal framework to understand individual documents and compare different time periods. It suggests that how climate change is framed, either as a wrongful act needing legal responsibility or as a shared problem requiring cooperation, affects the legal language in treaty texts. This language shapes the institutional setup and influences the types of funding methods available, such as mandatory fees or voluntary grants. These factors ultimately impact how predictable and sufficient financial support is for affected countries.



Figure 3.1: Five-Part Causal Framework for Analysing Evolution of Discourse

By looking at this framework over time, we can see how early decisions, like the 2015 exclusion of liability in a key decision, have influenced later developments, including the financing design at COP27 and COP28. The analysis also identifies key moments when the focus shifted from obligation to cooperation, changing how funding works. Additionally, it examines the gap between high-level ethical principles and practical financial mechanisms, revealing instances in which ineffective funding mechanisms weaken strong ideas.

Analytical Techniques

1. **Keyword Trends:** This method involves examining the discourse of liability-related terms compared to cooperative terms over specified time periods. The analysis not only examines the term but also explores its context, distinguishing between clause types.
2. **Thematic Clustering:** This approach categorises text into overarching themes based on the concepts it contains. It helps illustrate shifts in framing over time, facilitating an understanding of how certain issues are perceived and articulated.
3. **Instrument Evolution Table:** This method classifies financial proposals by source and obligation type. Examining various financial mechanisms enables a comparison between proposed and adopted instruments, providing insights into evolving approaches.
4. **Country-Country Matrix.** A structured matrix maps AOSIS, the LDC Group, the European Union, and the Umbrella Group across periods, tracking fund flows among countries and entities.

FINDINGS AND ANALYSIS

Historical Evolution of L&D Governance

The 1991 AOSIS Proposal

AOSIS has been calling for compensation since the early 1990s. During the negotiations of the UNFCCC, the group proposed the establishment of an international insurance scheme explaining how “[t]he resources of the insurance pool should be used to compensate the most vulnerable small island and low-lying coastal developing countries for L&D resulting from sea level rise” (Intergovernmental Negotiating Committee, 1991; UNFCCC, 1991b). It relied entirely on assessed, mandatory contributions from industrialised nations and calculated these exact obligations using a strict formula: nations paid 50% based on their gross national product, representing their economic capacity, and 50% based on their relative share of global carbon dioxide emissions, representing their historical culpability.

The mechanism consisted of three connected parts: (1) an Insurance Element; (2) a Remediation/Compensation Element; and (3) a Risk Management Element. The Compensation Element was considered essential for addressing the remaining L&D caused by gradual processes such as rising sea levels, warming land and ocean surface temperatures, and ocean acidification.

Notably, the proposal lists State Responsibility among its foundational concepts, alongside the Polluter Pays Principle, common but differentiated responsibilities and respective capabilities (CBDR-RC), the precautionary principle, principles of fairness and intergenerational fairness, and international unity. The inclusion of State Responsibility is particularly relevant here because it is fundamentally linked to the idea of compensation. Compensation is, in fact, a form of redress anticipated when State Responsibility arises from wrongful actions.

Furthermore, the pool was one of the first to advocate for direct compensation for permanent loss, openly acknowledging that mitigation and adaptation alone cannot reverse climate impacts.

The UNFCCC (1992) and the Kyoto Protocol (1997)

The 1992 UNFCCC, adopted at the Rio Earth Summit, dropped the assessed contribution pool entirely. Instead, Article 4.8 of the Convention survived as a political compromise: it merely

acknowledged the needs of vulnerable developing countries, listed insurance as an action that parties "shall give full consideration to," and provided no specific financial mechanism, replacing the demand for restitution with vague promises of future cooperation. Traces of this proposal can still be found in article 4.8 of the 1991 Convention, which calls on parties to consider appropriate actions, including "including (...) funding, insurance and the transfer of technology" (United Nations, 1992).

The 1997 Kyoto Protocol established binding greenhouse gas emission reduction targets exclusively for developed Annex I parties, centring the regime on mitigation (UNFCCC, 1997). As a result of the 1992 UNFCCC and the subsequent 1997 Kyoto Protocol, L&D was effectively buried as a distinct policy pillar, as developed nations feared endless compensation claims; the international climate regime focused strictly on preventing future emissions and adapting to expected changes.

The Marrakesh Accords (2001) and the Adaptation Funds

To facilitate adaptation, negotiators established specific financial mechanisms, such as the Special Climate Change Fund (SCCF) managed by the Global Environment Facility (GEF), established in 2001, which financed the additional costs of interventions measured against a predetermined development baseline (UNFCCC, 2001). By channelling climate finance exclusively through adaptation, the United Nations Framework Convention on Climate Change effectively declared all climate harms as adaptable conditions.

This institutional design choice left irreversible L&D without a dedicated financial mechanism. Funding adaptation inherently fails to cover the residual losses that occur when climate impacts breach physical and social adaptation limits, forcing SIDS and LDCs to absorb the costs of L&D during this period.

The Cancún Adaptation Framework (COP16, 2010)

The 2010 Cancún Adaptation Framework formally included L&D in a COP decision text for the very first time (UNFCCC, 2010). While it also created the first Loss and Damage Work Programme, it placed it under the adaptation umbrella and treated it as a management problem.

"13. Decides to hereby establish the Cancun Adaptation Framework encompassing the provisions laid out below, with the objective of enhancing action on adaptation, including through international cooperation and coherent consideration of matters relating to adaptation under the Convention;"

“25. Recognizes the need to strengthen international cooperation and expertise in order to understand and reduce L&D associated with the adverse effects of climate change, including impacts related to extreme weather events and slow onset events;”

“26. Decides to hereby establish a work programme in order to consider, including through workshops and expert meetings, as appropriate, approaches to address L&D associated with climate change impacts in developing countries that are particularly vulnerable to the adverse effects of climate change;”

- (UNFCCC, 2010, para. 13, 25-26)

It also explicitly avoids the issue and instead frames the issue as a need to "understand and reduce L&D", guiding the resulting work programme to focus on "risk management and reduction" alongside "risk sharing and transfer mechanisms such as insurance". Placing L&D inside the adaptation pillar locks the international response into a specific, restrictive path. If a treaty defines loss simply as an adaptation shortfall, it limits the available financial solutions to technical adjustments such as disaster risk reduction or micro-insurance.

Warsaw International Mechanism (COP19, 2013)

Leading up to COP19, vulnerable states faced escalating disaster costs. Existing climate finance failed to cover permanent destruction. Developing countries need a dedicated financial mechanism for irreversible climate impacts. Developed nations, on the other hand, feared unlimited financial exposure and suppressed treaty text establishing legal liability or mandatory compensation. This gridlock defined the political battleground.

The reality of this was demonstrated at COP19 when Super Typhoon Haiyan struck the Philippines, causing massive destruction, providing concrete evidence that mitigation and adaptation fail to prevent extreme losses (Santos et al., 2015). The Philippine negotiator Yeb Sano's hunger strike created intense global pressure, and developing country delegates and civil society observers walked out in protest of the industrialised nations' refusal to engage with the proposed financial mechanism, demonstrating that developing nations would no longer accept vague technical support in place of capital (Falzon et al., 2023; McGrath, 2013).

Developed nations, particularly the United States, the European Union, and the Umbrella Group, firmly opposed the establishment of a distinct financial mechanism due to domestic political constraints (Calliari et al., 2018, 2020). US negotiators operated under strict directives

to block any text that could create legal liability or require compensation and explicitly rejected liability and compensation while supporting technical help (Calliari et al., 2018, 2020).

To neutralise demands, developed countries used specific tactics as documented by Calliari (2014, 2018b) and Falzon (2023). First, they tried to trap L&D within existing adaptation and disaster risk reduction frameworks, arguing that current institutions were sufficient, as seen in the Cancun Frameworks. Second, they used scientific uncertainty to dispute liability, arguing attribution problems prevented direct compensation. Finally, they attempted to shift focus from financial restitution to non-economic losses and risk management strategies. Their financial suggestions focused on weak alternatives such as private insurance schemes, thereby shifting the burden onto vulnerable populations.

These incompatible positions created severe instability. Developing countries demanded a new institution with financial power. Developed nations refused to concede on liability and fought to restrict any new mechanism to information sharing and dialogue. This conflict forced an eleventh-hour deadlock, shaping the heavily compromised Warsaw International Mechanism.

Decision 2/CP.19 established the WIM with specific functions enhancing knowledge and understanding of comprehensive risk management approaches; dialogue, coordination, coherence, and synergies among relevant stakeholders; and action and support, including finance, technology, and capacity-building (UNFCCC, 2013, para. 5 (a), (b), (c)).

It grants the WIM authority to collect data and convene meetings but denies it the power to raise capital. While the third function mentions "finance", it mandates only the enhancement of support, not the provision of funds, establishing it strictly as a dialogue and knowledge mechanism, neutralising the demand for financial restitution.

Before Warsaw, the G77 and China submitted proposals demanding a dedicated mechanism comprising "means of rehabilitation and/or compensation" (G77 and China, 2013a, 2013b). The Alliance of Small Island States (AOSIS) demanded a financial facility with a compensatory component to address slow-onset events such as sea-level rise (AOSIS, 2012). Draft texts explicitly proposed a technical facility to redress unavoidable L&D alongside a financial facility to manage dedicated funding sources. Developed nations, led by the United States, meanwhile, actively deleted this vocabulary.

The final text of Decision 2/CP.19 gutted the proposed financial facility, leaving "only a work plan, without mention of a financial mechanism". "Compensation" and "liability" were entirely

expunged from the final agreement. Negotiators inserted vague directives to "enhance understanding" and promote "comprehensive risk management". Developing countries demanded a standalone L&D mechanism independent of adaptation. The final text placed the WIM "under the Cancun Adaptation Framework".

This structural choice legally subordinated L&D to adaptation. It framed permanent destruction as a future management problem rather than a present harm requiring redress. To force consensus, negotiators deployed contradictory language. The preamble of Decision 2/CP.19 acknowledges that L&D "involves more than that which can be reduced by adaptation", but the text locates the mechanism within the adaptation framework, allowing developed nations to accept the text without accepting liability (UNFCCC, 2013).

The Paris Agreement

The 2015 Paris Agreement created a new global climate framework and aimed to limit global warming to well below 2°C above pre-industrial levels (UNFCCC, 2015b). It rests on three specific pillars: mitigation to cut greenhouse gas emissions, adaptation to manage future risks, and L&D to address unavoidable climate impacts

While Article 8 of the Paris Agreement is a victory, as it establishes L&D as a pillar of climate action, the text itself fails to secure mandatory finance. A close reading of Article 8 reveals that developed nations replaced binding obligations with aspirational statements. Because the treaty relies on voluntary cooperation rather than legal duty, the system cannot generate the necessary revenue to repair irreversible climate harm.

Article 8 opens by establishing boundaries that protect developed nations from legal exposure. In Article 8.1, parties merely "recognize the importance of averting, minimizing and addressing L&D" (UNFCCC, 2015b). The verb "recognize" acknowledges it but demands no specific action. The verb "recognize" acknowledges a reality but demands no specific action.

Article 8.2 states the Warsaw International Mechanism (WIM) "shall be subject to the authority and guidance" of the parties (UNFCCC, 2015b, 2016b). While this clause uses the binding modal verb "shall", it only regulates the administrative reporting structure of the WIM. It delegates no fiscal authority and does not force any party to fund the mechanism.

The core evasion of financial responsibility occurs in Article 8.3, which states that parties "should enhance understanding, action and support... as appropriate, on a cooperative and facilitative basis" (UNFCCC, 2015b, 2016b). In legal drafting, the choice of a modal verb

determines the force of the law. Negotiators deliberately chose "should" rather than "shall". "Shall" creates a binding obligation. "Should" offers a recommendation. By choosing "should", the agreement reduces financial support to a discretionary political choice. Furthermore, the modifier "as appropriate" gives donor states an immediate exit, allowing them to withhold support whenever they deem it unsuitable. The phrase "cooperative and facilitative basis" reinforces this voluntary nature, legally severing the text from concepts of corrective justice or historical responsibility.

Article 8.4 defines the scope of this voluntary cooperation and states that areas of cooperation "may include" early warning systems, emergency preparedness, risk insurance facilities, and resilience building (UNFCCC, 2015b, 2016b). Again, the text uses the permissive modal "may" instead of a directive command. This list limits expected responses to technical risk management and insurance schemes that transfer the financial burden back to vulnerable populations. It contains no operative provision that compels wealthy nations to grant debt-free funds for permanent losses.

To ensure Article 8 created no financial exposure, negotiators drafted a specific exclusion clause outside the treaty text. Paragraph 51 of **Decision 1/CP.21** explicitly dictates that Article 8 "does not involve or provide a basis for any liability or compensation". This clause removes the juridical link between causation and compensation (UNFCCC, 2016b).

By eliminating liability, the agreement prevents vulnerable states from demanding reparations for historical emissions by industrialised nations. Consequently, Article 8 remains formally isolated from Article 9, which governs binding climate finance for mitigation and adaptation.

The discussions behind the Paris Agreement

By the 2015 Paris talks, the G77 and China bloc offered a major concession and drafted Article 8, omitting "liability" and "compensation" (Vanhala & Hestbaek, 2016). However, the United States rejected this compromise as lead US negotiator Todd Stern stated the US refused the notion of liability (Vidal, 2015). US Secretary of State John Kerry stated the US's reluctance as follows: "We are not against [L&D]. We're in favour of framing it in a way that doesn't create a legal remedy because Congress will never buy into an agreement that has something like that...the impact of it would be to kill the deal" (Neslen & Harvey, 2016).

To appease the US, the French COP Presidency released a late-stage draft that inserted a liability ban directly into the treaty text, mandating that L&D proceed "in a manner that does

not involve or provide a basis for liability or compensation" (Wewerinke-Singh & Salili, 2020). While Vanuatu recognised this trap and intervened during the final hours of the conference, Vanuatu's head of delegation called an emergency meeting with the Presidency. It threatened to block the consensus required to pass the Paris Agreement if the treaty retained the liability ban, halting the negotiations.

To break the deadlock, US delegates and small island leaders held closed-door meetings in which they abandoned the Presidency's draft treaty clause (Wewerinke-Singh & Salili, 2020). Instead, negotiators moved the liability exclusion out of the legally binding treaty and into the accompanying text as Paragraph 51 of Decision 1/CP.21.

Several small island states immediately exposed the political nature of this compromise. Upon signing the Paris Agreement, Vanuatu and others filed formal declarations stating their ratification did not constitute a renunciation of any rights under international law. By shifting the text to the decision document, developed nations secured their financial immunity, but developing nations kept the treaty text free of a permanent legal ban.

Developments after the Paris Agreement leading to 2022

In 2017, the WIM adopted its five-year rolling work plan but excluded new public finance, steering vulnerable countries toward private insurance and risk pooling (UNFCCC, 2017). High-emitting states replaced specific mandates with dialogues and commercial risk products, forcing poor nations to pay market premiums for their own protection while shielding wealthy nations from liability (Gewirtzman et al., 2018).

This approach failed on a scale comparable to that of actual disasters when Hurricane Maria struck Dominica in 2017, causing \$1.37 billion in damage. Yet, the regional insurance facility paid out only \$19.3 million (Richards & Schalatek, 2017).

Negotiators then offered technical advice rather than capital to delay binding commitments. At COP25 in 2019, parties established the Santiago Network to connect vulnerable countries with technical experts (UNFCCC, 2019). However, this body provided no new money to repair irreversible harm, focusing entirely on sharing knowledge and risk management strategies.

The refusal to provide concrete funding reached a breaking point at COP26 in 2021. The G77 and China bloc demanded the creation of a Glasgow Loss and Damage Facility to disperse funds directly (Falzon et al., 2023; McDonnell, 2023).

The United States, Australia, and the European Union rejected this proposal and offered the Glasgow Dialogue, a three-year discussion process with no mandate to secure actual funding. This tactic extended the period of talk without action, leaving vulnerable states without financial recourse as global temperatures continued to rise (Falzon et al., 2023; McDonnell, 2023).

Developments at COP27

With the failure of the Glasgow Dialogue coinciding with escalating climate catastrophes, such as the 2022 Pakistan floods, COP27 at Sharm el-Sheikh forced developing countries to accept L&D finance onto the official agenda after overnight negotiations (Adger, 2023; Falzon et al., 2023; McDonnell, 2023).

Under intense pressure, developed nations conceded. Parties agreed to establish a dedicated Loss and Damage Fund. Although this decision ended decades of obstruction, the agreement leaves the exact funding sources unresolved, keeping the actual flow of money uncertain (Adger, 2023; Falzon et al., 2023; McDonnell, 2023).

At COP27 in Sharm el-Sheikh, parties agreed to establish new funding arrangements and a dedicated fund to assist particularly vulnerable developing countries affected by climate change, acknowledging an "urgent and immediate need for new, additional, predictable and adequate financial resources" to respond to both economic and non-economic harm (UNFCCC, 2023). However, the text serves only as an empty vessel, lacking operative provisions that define who pays, how much they pay, and the specific mechanisms that will generate revenue.

The COP27 decision deliberately does not mandate assessed contributions based on historical emissions or economic capacity, as developed nations refused to link the new fund to liability or compensation, making all financial transfers remain voluntary (Falzon et al., 2023; McDonnell, 2023; Siegele, 2023; UNFCCC, 2023).

Furthermore, the text broadens the scope of support to a "mosaic" of funding arrangements. This framing allows wealthy nations to classify existing humanitarian aid, loans, or private insurance schemes as L&D finance, thereby fragmenting the architecture and undermining the demand for a unified, grant-based mechanism.

To address these missing financial and structural elements, the decision delegates the design of the fund to a 24-member Transitional Committee, which will define the institutional

arrangements, modalities, governance, and terms of reference (UNFCCC, 2023). Crucially, the committee must identify and expand on specific funding sources.

Developments at COP28

At COP28 in Dubai, parties adopted the Governing Instrument for the Fund for Responding to Loss and Damage (FRLD), marking a historic step toward providing financial assistance to developing nations highly vulnerable to climate change impacts (UNFCCC, 2024). The fund, operating under the UNFCCC and the Paris Agreement, is designed to address both economic and non-economic losses, including those from extreme weather and slow-onset events and is directed to fill priority gaps within the existing finance architecture (UNFCCC, 2024).

It establishes a 26-member Board to govern the money. It sets the rules for a new, independent secretariat, with the World Bank hosting the secretariat and serving as the interim trustee for four years (Board of the FRLD, 2024a; Board of the FRLD & IBRD, 2024). While developing nations wanted an independent institution, developed nations wanted World Bank control. To resolve this, the rules force the World Bank to grant direct access to developing countries and allow the Fund's rules to supersede Bank policies if they clash (Board of the FRLD, 2024a).

The summit also secured the first financial commitments. Donor countries pledged USD 792 million to the broader L&D funding arrangements, directing USD 661 million specifically to the FRLD (Board of the FRLD, 2024c). Germany, Italy, France, and the UAE led these initial pledges, each offering USD 100 million or more (Board of the FRLD, 2024c, 2024b). Additionally, Japan provided USD 10 million directly to the UNFCCC to cover the interim secretariat's setup costs (Board of the FRLD, 2024c). While these pledges allowed it to exist on paper, they accounted for less than 1 per cent of the hundreds of billions needed annually, leaving an unresolved gap between the Fund's goals and its resources (Heinrichs et al., 2025).

Developments at COP29

COP29 in Baku shifted the focus from creation to legal accountability as the conference approved formal arrangements between the COP, the CMA, and the FRLD Board (UNFCCC, 2025a). During this summit, the Board signed a Host Country Agreement with the Philippines, granting it independent legal personality and the capacity to negotiate contracts. By securing legal standing, the Board removed the final administrative barrier to hiring an Executive Director and forming the permanent secretariat.

The Baku summit also exposed the slow pace of revenue generation, with the COP issuing its first formal guidance to the Fund, urging donor countries to convert their pledges into signed contribution agreements quickly (UNFCCC, 2025a). Pledges grew marginally to approximately USD 820 million, aided by new commitments from Luxembourg, Spain, and the Walloon Region of Belgium (UNFCCC, 2025a).

To unite the fragmented L&D network, the FRLD co-convened the first annual high-level dialogue at the World Leaders Climate Action Summit, bringing together multilateral banks, United Nations agencies, and state leaders to align their scattered funding tools (Board of the FRLD, 2024d).

Developments at COP30

At COP30 in Belém, the FRLD transitioned from planning to active funding, launching the first call for funding requests on the opening day of the summit (Board of the FRLD, 2026c, 2026b). This call operates under the Barbados Implementation Modalities (BIM). This short-term testing phase allocates USD 250 million across 2025 and 2026, distributing grants ranging from USD 5 million to USD 20 million to help vulnerable countries test local interventions and build national response systems (Board of the FRLD, 2025d, 2025a, 2026b). To protect the most exposed nations, the Board guaranteed that SIDS and LDCs would receive at least 50 per cent of this initial money.

The Belém summit also tested the Fund's ability to coordinate with external agencies, deploying a dedicated team to support the negotiations directly. To demonstrate that the Fund does not operate in isolation, the FRLD co-hosted an event with the Santiago Network and the Warsaw International Mechanism Executive Committee, showcasing how the three pillars of the L&D architecture (knowledge, technical assistance, and finance) combine to support developing countries (Board of the FRLD, 2026c). The call for funding requests remains open until 15 June 2026, marking the first time vulnerable states can access dedicated multilateral cash to rebuild after climate disasters (Board of the FRLD, 2026a).

Analysing FRLD in detail

The Governing Instrument gives the FRLD a broadly specific mandate, authorising the Fund to assist particularly vulnerable developing countries in responding to climate change (UNFCCC, 2024). While the mandate covers both economic and non-economic loss and damage, it explicitly targets impacts from extreme weather events and slow-onset events

(UNFCCC, 2024). The text directs the Fund to support actions that complement immediate humanitarian relief, focusing instead on intermediate or long-term recovery, reconstruction, and rehabilitation (UNFCCC, 2024). The Fund also supports responses to climate-induced human mobility, including displacement and planned relocation (UNFCCC, 2024).

Paragraph 2 states that the Fund exists to assist developing countries that are particularly vulnerable to climate change in responding to economic and non-economic L&D, while Paragraphs 8 and 9 expand this scope to cover recovery, reconstruction, slow-onset events, and climate-induced displacement (UNFCCC, 2024). The Governing Instrument acknowledges that communities lose lives, cultural heritage, and territory, but doesn't provide a corresponding mechanism to compel historic polluters to pay for the losses.

To finance these operations, the Governing Instrument permits the Board to receive funds from a "wide variety of sources of funding, including grants and concessional loans from public, private and innovative sources. Still, it sets no minimum capitalisation requirement (UNFCCC, 2024). The accompanying **decision 1/CP.28** urges developed countries to provide support, but explicitly frames these contributions as voluntary (UNFCCC, 2024). Contributors deposit payments in cash or promissory notes via individual contribution agreements, retaining complete discretion over funding volume and frequency. It also entirely omits mandatory financial tools, defining no formula for assessed contributions, setting no legally binding capitalisation target, and providing no mechanism for levies, thereby leaving the FRLD without an enforcement mechanism to penalise non-contributing parties.

The institutional design further limits the FRLD's autonomy, as **Decisions 1/CP.28 and 5/CMA.5** appoint the World Bank as the interim host and trustee for 4 years (UNFCCC, 2024). But while the Governing Instrument supersedes World Bank policies during disputes, the FRLD must bear the World Bank's direct and indirect cost-recovery fees, subjecting the Fund to the fiduciary architecture of an existing multilateral bank rather than establishing an independent financial entity with its own revenue-raising powers. (Board of the FRLD, 2025b, 2025b; Board of the FRLD & IBRD, 2024).

The agreement also locks in these constraints for years. Paragraph 55 mandates a periodic replenishment every four years, while paragraph 66 subjects the Fund to periodic reviews conducted by the COP and the CMA based on independent evaluations (UNFCCC, 2024). In addition, the COP will assess the World Bank's performance at the end of the four-year interim period before deciding whether to establish the Fund as an independent stand-alone institution

and serve as the only formal mechanism for vulnerable states to contest the Fund's design, guaranteeing that developed nations face no immediate pressure to alter the voluntary model. (Board of the FRLD & IBRD, 2024; UNFCCC, 2024).

Years	Milestones	L&D Finance Advocacy	L&D Finance Obstructions
1991-2007	1991: Vanuatu, on behalf of AOSIS, proposed an international insurance for sea level rise. 1992: Adoption of the UNFCCC. Insurance (by AOSIS) is listed under Article 4.8 only as a meeting option. 2001: Marrakesh Accords established LDCF and SCCF. 2007: Bali Action Plan Decision 1/CP.13 formally included L&D terminology in negotiation text.	Vanuatu and AOSIS pushed for L&D inclusion and proposed a collective-loss-sharing insurance scheme to compensate victims of sea-level rise. They framed L&D as adverse effects and risk transfer. They argued for funding based on historical responsibility, emissions and capacity GNP. They pushed for new, additional, and adequate revenue.	Developed nations excluded the specific insurance proposal from final agreements and the 1992 Convention. No funding mechanism was established under Article 4.8. The article conflated L&D with the impacts of response measures for oil producers. Global North shifted discussions to informal workshops and technical programs to avoid financial commitments.
2008-2013	2008: AOSIS submitted a multi-window mechanism proposal including insurance, rehabilitation and compensation, and risk management. 2010: Cancun Agreements established a work program on L&D under the Cancun Adaptation Framework. 2012: Doha COP18 Decision 3/CP.18 moved toward institutional arrangements. 2013: Warsaw International Mechanism (WIM) launched in COP19.	AOSIS, LDCs, and the African Group coalition, later unified as the G77 plus China, proposed compensation and rehabilitation mechanisms. They demanded strong negotiation texts in Warsaw on a dedicated institutional body and financial assistance to address unavoidable impacts. A Philippine negotiator's 2013 hunger strike drew global media attention to the demand for a solidarity fund.	Developed nations, led by the US and UK, weakened the WIM language to ensure it remained dialogue-based and restricted to knowledge sharing. They forced the WIM to be established under the Cancun Adaptation Framework to restrict its scope and avoid separate finance tracks. The UK cut specific funding support for L&D-related initiatives. They conflated L&D with adaptation to deflect from historical responsibility.
2014-2015	2014: WIM's initial two-year work plan was adopted. 2015: Paris Agreement Article 8 established L&D as a standalone concept or pillar. 2015: Decision 1/CP.21 Paragraphs 51 and 52 accompanied the Paris Agreement.	Developing nations secured recognition of L&D as a third pillar distinct from Article 7 adaptation. AOSIS 1.5 degrees C to stay alive campaign linked L&D to ambitious temperature goals. They broadened the framing to include non-economic losses such as life, culture, and identity, as well as migration and displacement. NGO and media pressure increased around climate justice.	The US and Australia resisted a standalone status until the final hours and blocked linkage between Article 8 and mandatory finance. Negotiators forced the inclusion of Paragraphs 51 and 52, which explicitly excluded liability or compensation from Article 8. Efforts were made to steer L&D finance toward private insurance and pilot risk transfer schemes rather than public grants.
2016-2022	2016: First WIM review held in Marrakesh. 2017: WIM five-year rolling workplan and Fiji Clearing House for Risk Transfer established. 2019: Santiago Network for L&D established at COP25. 2021: The Glasgow Dialogue was established after developed nations blocked a finance facility at COP26. 2022: COP27 Decision 2/CP.27 established the Fund for responding to L&D FRLD.	G77 plus China unified to demand a dedicated L&D Finance Facility and for L&D to be a standalone agenda item. They secured inclusion of L&D in the Global Stocktake. They pushed for new and additional financing beyond ODA. The Scottish Government made the first voluntary commitment in 2021. The V20 group advocated for regional risk pools and InsuResilience.	Developed nations, including the US, EU, and Australia, initially blocked the finance facility proposal in Glasgow and instead offered the non-binding Glasgow Dialogue. They promoted insurance-based alternatives such as Global Shield and InsuResilience. They delayed funding decisions through technical work programs and slowed operationalisation of the Santiago Network. They pushed for a broad donor base to include emerging economies to dilute historical responsibility.
2023- present	2023: Transitional Committee meetings finalised fund design, and FRLD was operationalised at COP28. 2024: Board Meetings B1 to B4 held. Ibrahima Cheikh Diong appointed	G77 plus China secured institutionalisation of the fund and pushed for grant-based, predictable, new, and additional finance. They pushed for direct access for subnational and national entities, direct	The Global North insisted on the World Bank as the interim host and trustee despite concerns about autonomy and bureaucracy. They preferred voluntary contributions over assessed obligations based on historical emissions.

	Executive Director. The Philippines was selected as the host country. 2024: Pledges mobilised for FRLD, and the Trustee Agreement was signed with the World Bank. 2025: First Annual High-Level Dialogue scheduled and Barbados Implementation Modalities launched. 2026: Expected first funding round and initial project approvals.	budget support, and community-led funding. SIDS and LDCs secured a 50 per cent allocation floor in the Barbados Implementation Modalities. Advocacy focused on rapid delivery and country-led interventions.	They continued to resist liability and compensation, as well as human rights-based legal framing. Initial pledges remained limited relative to projected needs in the hundreds of billions of dollars.
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Table 4.1: Five Periods of Contention over Loss and Damage Finance in the UNFCCC and Positions Taken by Advocates and Opponents, inspired by Falzon et al. (2023).

Country-Country Shifts in Opinion

The consensus process of global climate governance forces vulnerable states to trade legal rights for institutional survival. While developing nations seek corrective justice for permanent climate destruction, wealthy nations refuse to pay. To prevent negotiations from collapsing, vulnerable states consistently abandon their demands for mandatory compensation. This dynamic produces empty institutions that lack fiscal authority.

Negotiating Bloc	Phase I 1991–2007	Phase II 2008–2013	Phase III 2014–2015	Phase IV 2016–2022	Phase V 2023–present
AOSIS Alliance of Small Island States	L	L	L	L	C
LDC Group Least Developed Countries	L	L	L	L	C
European Union EU Member States	HO	HO	SO	SO	SO
Umbrella Group US, Australia, Canada, Japan	HO	HO	SO	SO	SO

L Liability C Cooperation SO Soft Opposition HO Hard Opposition N Neutral T Transitions

Figure 4.1: Country-Coalition Position Matrix (1991–present) tracking how the language of L&D governance shifted among key negotiating blocs.

The matrix reveals a clear trajectory of "demand migration" among the developing nations, as the AOSIS and the LDC Group moved from demanding strict liability to accepting frameworks built entirely on voluntary cooperation. Vanuatu, on behalf of AOSIS, initially demanded a mandatory, emissions-based insurance mechanism funded by wealthy nations, consistently framing it as corrective justice until 2012 (Calliari, 2018b; Intergovernmental Negotiating Committee, 1991). However, by 2023, these same blocs accepted the FRLD, moving from a demand for mandatory, liability-based funding to an institution that relies exclusively on voluntary pledges and explicitly rejects liability (Falzon et al., 2023).

Yet, this migration does not mean their normative beliefs changed. Throughout the 33 years, submissions from AOSIS and the LDCs continue to invoke historical responsibility and climate justice. Instead, their strategic calculus shifted (Wewerinke-Singh, 2023). Negotiators from vulnerable states calculated that securing a formal mechanism with weak fiscal authority was better than securing no mechanism at all (Calliari, 2018b; Wewerinke-Singh & Salili, 2020).

The primary inflexion point occurred at COP19 in 2013, when AOSIS and the LDC Group entered the Warsaw negotiations, demanding a mechanism with a clear mandate for compensation (Calliari, 2018b). Developed nations, particularly the United States, outright rejected this, leading developing nations to accept the WIM, a body stripped of authority and restricted to sharing knowledge (Calliari, 2018b).

Before 2013, the European Union and the Umbrella Group used hard opposition, rejecting L&D as a distinct category and forcing all discussions into the adaptation framework to avoid any implication of liability (Calliari, 2018b). After COP19, their strategy evolved into "soft opposition, as they now accept the L&D label but actively strip the resulting mechanisms of any financial obligation (Falzon et al., 2023). By replacing clear legal duties with ambiguous text regarding "support" and "dialogue," they appear cooperative while avoiding actual payments.

This soft opposition defines the inflexion point between COP26 and COP27. At COP26 in 2021, the G77 and China demanded a Glasgow Loss and Damage Facility to disburse funds directly, but this was blocked, with the alternative being the "Glasgow Dialogue" (Falzon et al., 2023). By COP27, escalating physical disasters, such as the 2022 Pakistan floods, combined with intense pressure from the G77, forced the EU and Umbrella Group to shift tactics and agree to establish a dedicated fund. However, they immediately applied their soft opposition strategy, ensuring the COP27 decision explicitly separated the new fund from liability or compensation, allowing the FRLD to exist but functioning as a mechanism for voluntary charity rather than a tool for corrective justice (Falzon et al., 2023).

Evolution of Instruments Used

Existing financial mechanisms fail in the face of the physical realities of permanent climate destruction. When infrastructure washes away, or land turns to desert, the resulting intervention generates no financial return.

Instrument	Mechanism	L&D Application & Political Status
Grants	Direct cash transfers require no repayment.	Status: Adopted. Designated as the primary instrument for the start-up phase of the new FRLD (Board of the FRLD, 2025c). Developing nations demand grants to avoid sovereign debt.
Concessional Loans	Debt issued at below-market interest rates.	Status: Adopted but delayed. Included in the FRLD Governing Instrument for future use (Board of the FRLD, 2025c). Vulnerable states oppose loans because they force victims to borrow money to recover from disasters, they did not cause, driving them into debt crises.
Insurance & Risk Pooling	Policyholders pay premiums to transfer risk to private insurers or regional pools.	Status: Adopted and heavily promoted by developed nations. Fails for inevitable slow-onset events and cannot value NELD
Catastrophe Bonds	High-yield debt instruments that forgive principal if a specific disaster trigger is met.	Status: Adopted in limited cases. Provides rapid liquidity but incurs high fixed costs and covers only sudden, unpredictable catastrophes, ignoring slow-onset hazards.
Contingency / Pre-arranged Finance	Pre-approved credit lines that governments draw from immediately after a disaster.	Status: Adopted. Offers rapid post-disaster liquidity but ultimately adds to sovereign debt burdens.
Direct Budget Support	Financial transfers are sent directly to national treasuries to fund public policy or emergency response.	Status: Adopted. Listed in the FRLD Governing Instrument. Allows rapid, country-led disbursement following extreme weather events.
Levies and Taxes	Mandatory surcharges on international airline travel, bunker fuels, financial transactions, or fossil fuel extraction.	Status: Proposed but stalled. Offers predictable, new revenue aligned with the polluter-pays principle. High-emitting nations consistently block implementation.
Solidarity Funds	Pools of money are built from voluntary pledges by wealthy nations, regional governments, or philanthropies.	Status: Adopted. Operates outside strict liability. Relies entirely on discretionary charity, which makes revenue unpredictable and usually inadequate.
Compensation Facilities	Mandatory payments are extracted from nations based on their historical emissions and wealth.	Status: Rejected. Developed nations strictly veto any mechanism that admits legal liability or mandates restitution.
Equity Investments	Capital is provided in exchange for ownership shares and expected financial returns.	Status: Authorised in FRLD but practically useless. Submerged land and destroyed infrastructure generate no dividends, leaving equity investors with no underlying asset to hold.

Table 4.2: Every major financial mechanism discussed within the United Nations Framework Convention on Climate Change (UNFCCC).

Yet, developed nations consistently promote market-based instruments and debt-generating loans to fund L&D, while vulnerable states demand debt-free grants and direct levies on polluters. This structural conflict defines the complete inventory of proposed, active, and rejected L&D financial instruments.

Evaluating only adopted climate agreements hides how negotiators dismantle binding financial mechanisms. To understand the actual trajectory of L&D governance, one must examine the proposals that failed. When analysing the rejected instruments, the historical pattern of climate finance negotiations reveals a consistent strategy by wealthy nations to reject mandatory obligations, leading to a continuous weakening of ambition in successive proposals.

Year	Proposing party	Instrument type	Obligation	Outcome	Grounds for rejection or dilution
1991	AOSIS	International insurance pool with assessed contributions (50% by GNP, 50% by \$CO ₂ share)	Mandatory	Rejected	Liability concerns led to being dropped entirely from the UNFCCC in 1992. Replaced by vague Article 4.8 recognition with no financial mechanism.
2001	AOSIS / G77	Compensation and rehabilitation fund for irreversible climate impacts	Mandatory	Rejected	Subsumed into the Special Climate Change Fund (SCCF) as an adaptation instrument, the compensation element was removed.
2008	AOSIS	Multi-window mechanism: insurance element + compensation and rehabilitation element + risk management element	Mandatory	Rejected	Donor opposition; Warsaw International Mechanism (2013) substituted, dialogue and knowledge body only, no financial authority.
2013	G77 and China	Dedicated financial facility with a compensatory component for unavoidable and slow-onset impacts	Mandatory	Rejected	Final Decision 2/CP.19 stripped the financial mechanism entirely. "Compensation" and "liability" expunged. WIM is confined to knowledge sharing.
2021	G77 and China	Glasgow Loss and Damage Facility — a dedicated fund to disburse grants to affected countries	Mandatory	Rejected	The US, EU, and Australia blocked the facility. Replaced with non-binding Glasgow Dialogue — a three-year discussion process with no funding mandate.

2023	AOSIS / LDC Group	Independent FRLD with mandatory capitalisation targets, assessed contributions, and levy mechanisms.	Mandatory	Partially rejected	World Bank hosting adopted over independence; voluntary contributions only; no levy mechanism; no capitalisation floor. The Governing Instrument excludes all mandatory revenue authorities.
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Table 4.3: Every major financial instrument proposed during L&D negotiations that final decision texts either rejected or neutralised. A detailed table of all instruments will be included in the Annexures.

From vetoing AOSIS's 1991 demand for a fund based on assessed contributions to removing mandatory revenue mechanisms from later demands for compensatory and funded facilities, developed nations have consistently favoured voluntary rather than binding duties. This deliberate rejection has resulted in architecturally weaker instruments, shifting from a structured, compensatory pool to mechanisms with no financial component, and ultimately to the FRLD in 2022-23, which relies entirely on voluntary pledges.

Furthermore, a "substitution strategy" is employed, in which demands for financial mechanisms are deflected and replaced with non-financial alternatives such as knowledge-sharing committees, procedural delays, or technical assistance, effectively substituting capital demands with research mandates and advice.

Current Financial Status of FRLD

The December 2025 Trustee Report (2025), the first annual financial statement of the FRLD, provides the clearest possible window into the gap between the fund's stated purpose and its operational reality. Total pledges and contributions to the FRLD as of December 31, 2025, stand at USD. 828 million, of which USD 456 million has been received in cash (World Bank, 2025). This figure, presented as a headline achievement, requires immediate contextualisation. Conservative academic estimates place annual loss and damage costs for developing nations at \$400 billion by 2030, rising steeply thereafter (Markandya & González-Eguino, 2019). The FRLD's total potential resources represent approximately 0.1% of a single year's projected need.

Category	USD. (millions)
Cash actually received	456.75
Contributions signed but unpaid	144.79
Pledges (expressions of intent only)	226.92
Total (as of December 31, 2025)	828.47
Total funds ready for usage	420.79

Nearly half the headline number (\$371.71 million) consists of resources not yet received, a substantial portion of which are pledges that carry no binding legal obligation. The substitution strategy documented in rejected proposals from 1991 onwards has, in this iteration, entered the capitalisation architecture itself: headline numbers are announced to signal ambition, while the underlying legal structure permits non-delivery.

Contributor-Level Analysis



Figure 4.2: The Treemap represents total amount pledged by contributors (USD). The fill intensity represents the percentage of funds actually delivered. Countries that have delivered 0% are highlighted with a red outline and fill. Data: World Bank FRLD Trust Fund Trustee Report, December 31, 2025.

Italy pledged EUR 100 million (USD. 117.47 million). The European Commission pledged EUR 25 million (USD. 29.37 million), none received as of December 31, 2025 while the UAE, which made its pledge during COP28 on its home territory, has transferred only USD 15 million of a USD 100 million pledge, leaving USD 85 million outstanding (World Bank, 2025).

The United States, historically the primary blocker of mandatory L&D finance (Calliari, 2018b; Falzon et al., 2023; McDonnell, 2023), contributed USD 17.56 million, reflecting both its constrained political willingness and the absence of any assessed contribution mechanism that would have required a proportionate share based on historical emissions or GDP. Under the 1991 AOSIS formula (50% by GNP, 50% by CO₂ share), the US obligation would have been orders of magnitude higher. With the FRLD Board deciding to set aside USD 250 million for the Barbados Implementation Modalities call for funding requests, pending subsequent funding decisions, there is an urgent requirement for more funds to be paid out for the FRLD.

Contributor	Curr	Pledges in Currency of Contribution	USDeq. c/	Effective (or signed) Contribution in Currency of Contribution	Receipts in Currency of Contribution	Cash Receipts in USD d/ e/	Unencashed Promissory Notes in USDeq. f/	Cash Receipts in USD + Unencashed Promissory Notes in USDeq.	Total Outstanding Pledges in USDeq. f/	Total Outstanding Contributions in USDeq. f/
Australia	AUD	50.00	33.14	50.00	17.00	11.08	-	11.08	-	22.06
Austria	EUR	25.00	27.64	25.00	15.00	15.90	-	15.90	-	11.75
Belgium (Walloon Region)	EUR	3.00	3.40	2.00	2.00	2.23	-	2.23	1.17	-
Canada	CAD	16.00	11.63	16.00	16.00	11.63	-	11.63	-	-
Denmark	DKK	175.00	24.66	175.00	175.00	24.66	-	24.66	-	-
Estonia	EUR	0.05	0.05	0.05	0.05	0.05	-	0.05	-	-
European Commission	EUR	25.00	29.37	-	-	-	-	-	29.37	-
Finland	EUR	3.00	3.52	3.00	3.00	3.52	-	3.52	-	-
France	EUR	100.00	111.81	50.00	50.00	53.07	-	53.07	58.74	-
Germany	EUR	92.00	98.24	92.00	92.00	98.24	-	98.24	-	-
Iceland	USD	0.60	0.60	0.60	0.60	0.60	-	0.60	-	-
Iceland	ISK	320.00	2.55	-	-	-	-	-	2.55	-
Ireland	EUR	25.00	28.10	25.00	25.00	28.10	-	28.10	-	-
Italy	EUR	100.00	117.47	-	-	-	-	-	117.47	-
Japan a/	USD	15.00	15.00	15.00	15.00	15.00	-	15.00	-	-
Korea	USD	7.00	7.00	7.00	2.15	2.15	-	2.15	-	4.85
Latvia	EUR	0.05	0.05	0.05	0.05	0.05	-	0.05	-	-
Luxembourg	EUR	15.00	17.62	-	-	-	-	-	17.62	-
Netherlands	EUR	15.00	15.78	15.00	15.00	15.78	-	15.78	-	-
New Zealand	NZD	10.00	5.66	10.00	10.00	5.66	-	5.66	-	-
Norway	USD	25.00	25.00	25.00	25.00	25.00	-	25.00	-	-
Portugal	EUR	5.00	5.69	5.00	2.00	2.16	-	2.16	-	3.52
Slovenia	EUR	1.50	1.73	1.50	0.40	0.44	-	0.44	-	1.29
Spain	EUR	40.00	44.01	40.00	40.00	44.01	-	44.01	-	-
Sweden	SEK	200.00	21.64	200.00	50.00	5.33	-	5.33	-	16.32
United Arab Emirates	USD	100.00	100.00	100.00	15.00	15.00	-	15.00	-	85.00
United Kingdom b/	GBP	40.00	53.26	40.00	40.00	26.35	26.91	53.26	-	-
United States	USD	17.56	17.56	17.56	17.56	17.56	-	17.56	-	-
Total			822.19			413.57	26.91	440.48	226.92	144.79

Table 4.4: Contributor breakdown. Promissory note contributions and all non-USD amounts are valued based on exchange rates as of December 31, 2025. Note that totals may not add up due to rounding.

The World Bank Hosting Problem, Expressed in Data

The FRLD's assets are managed not by an independent climate body but as part of the World Bank's commingled investment portfolio. The undisbursed cash balance is maintained in a commingled investment portfolio for all trust funds managed by IBRD, with funds managed such that the expected maximum loss in the worst 1% of cases will not exceed 0.25% at the portfolio's investment horizon (World Bank, 2025).

From inception through December 31, 2025, the FRLD Trust Fund earned investment income of approximately USD 16.28 million, reflecting a portfolio return of 4.96% (World Bank, 2025). It capitalised on the principle that polluters owe a debt to climate victims, generating returns by investing in the same financial system that finances the fossil fuel economy. Whether those holdings include fossil fuel bonds or not, the fund earns money by sitting idle while communities it was created to serve experience irreversible losses.

The UK's contribution (GBP 40 million delivered as a promissory note rather than cash) creates a USD 4.04 million currency risk reserve that directly reduces funds available for FRLD Board decisions (World Bank, 2025). The World Bank's risk management conventions, designed for a broad portfolio of development lending, draw on project-eligible resources from a fund intended for victims of climate disasters.

DISCUSSION

The evidence in the findings produces one answer to the primary research question. The normative shift in UNFCCC texts from liability to cooperation structurally limits the FRLD's authority to enforce mandatory revenue measures. The distinction between a structural constraint and a political failure is not semantic, as political failure is reversible. If the FRLD's inadequacy came from weak leadership, poor advocacy, or an unfavourable political climate in major donor countries, then better negotiators or a change of government could fix it. Structural constraints work differently. They are enshrined in the institution's founding documents and do not yield to political goodwill. Changing them requires amending the architecture, which means reopening the agreed text.

The FRLD Governing Instrument confirms that this is a structural problem. Paragraph 2 authorises the Fund to assist particularly vulnerable developing countries, but it sets no minimum capitalisation requirement and defines no formula for assessed contributions (UNFCCC, 2024). Decision 1/CP.28 explicitly frames donor contributions as voluntary (UNFCCC, 2024). The Fund receives money through individual contribution agreements that give donors full discretion over volume and timing. The FRLD cannot require mandatory contributions not because its leadership has failed to ask for them, but because its founding document does not authorise it to do so.

The fiscal consequence is measurable. Conservative projections place annual loss and damage costs for developing nations at USD 290-580 billion by 2030, rising to USD 1.1-1.7 trillion by 2050 (Markandya & González-Eguino, 2019). Total pledges to the FRLD as of December 2025 stood at USD 822 million, of which USD 413 million had been received in cash (World Bank, 2025). The Fund's current capital is less than 0.2% of the projected need, a structural gap that will persist due to its lack of mandatory revenue authority.

The policy implication follows directly. Incremental improvements to the FRLD's governance, transparency, or operational efficiency will not resolve the funding gap. Only two pathways can: redesigning the mechanism to include some form of mandatory obligation, or building parallel mechanisms outside the FRLD on different legal foundations (Wewerinke-Singh and Salili, 2020).

Constructive Ambiguity as Destructive Ambiguity

The destructive ambiguity concept extends Vanhala's (2023) constructive ambiguity analysis. Vanhala's governance study identified fiscal consequences as a concern, but did not develop the mechanism in institutional economics terms.

This concept corrects the usual, often praised, scholarly view of constructive ambiguity in environmental agreements. While the standard tale stresses the helpfulness of vague treaty words in securing written agreement, analysis of L&D Governance shows a worse institutional outcome. Specifically, when the main point of financial duty or money-raising power is made deliberately unclear, the result is not just putting off hard choices but a fixed bias towards the least costly standard.

This structural lack of money happens because the tougher standard, namely payment or duty, always needs clear positive legal writing to work. The simpler, voluntary standard (risk control and willing collaboration) is set up just by having no such clear writing. The quiet space made by unclear wording consistently helps the party wanting to lessen their duty, turning a useful negotiating tool into a fixed financial limit. The big goal in the FRLD's task, to deal with yearly losses in the hundreds of billions, is completely ruined by its own basic words. This destructive ambiguity thus gets written agreement but leaves the resulting body financially weak, a key insight for both climate governance and institutional design theory. It demands a fresh look at how soft-law rules are viewed when the subject is funding rather than programme range, suggesting that in these cases, unclear wording does not allow slow rule growth but permanently blocks the higher-cost option.

The Free-Rider Problem

Big past emitters who made no promise face no penalty now, as no penalty system exists (Falzon et al., 2023). Åberg and Jeffs (2022) show that key funders of global climate schemes often make large promises but pay out only part of them. Van Schie et al. (2024) demonstrate that these gaps are natural in voluntary finance, not just the result of a few bad actors.

The 1991 AOSIS rule offers a clear sharing guide: 50 per cent by gross national product and 50 per cent by CO₂ share (Intergovernmental Negotiating Committee, 1991). Under any such rule, the United States would owe the most, as it is both the largest total emitter and the biggest economy.

Yet, the US pledged only USD 17.56 million to the FRLD by December 2025, showing both weak political will and a complete lack of any firm payment system (World Bank, 2025). The same pattern as other big donors supports the idea, with Italy promising EUR 100 million and giving nothing by year-end 2025. The European Commission promised EUR 25 million and gave nothing. With such

Pathways Forward

This thesis tracked the evolution of Loss and Damage governance under the UNFCCC from 1991 to 2024, using four analytical methods: keyword discourse analysis across the full corpus of L&D documents; a Country-Coalition Position Matrix mapping negotiating positions over time; an Instrument Evolution Table cataloguing every major financial proposal; and a clause-level fiscal authority assessment of the FRLD Governing Instrument.

Under current rules, solidarity levies on global flights or ships look best. These charges are for activity, not state responsibility. They dodge the liability problems in Decision 1/CP.21, paragraph 51. They could quickly send new money to the FRLD without a party admitting legal fault (Pill, 2022). Similar ideas have often stalled in UNFCCC talks for political, not legal, reasons. A "solidarity" view may work better than past "polluter-pays" ideas, which rich nations see as a direct liability claim.

Multilateral Development Bank (MDB) channelling offers a short, partial fix. MDB climate money for Loss and Damage (L&D) work could briefly close the funding gap. The core snag is that MDB funds are mostly loans (CPI, 2023). As Adeniyi (2025) confirms, borrowing by poor nations to recover from disasters they did not cause is wrong and creates sovereign debt crises. The difference between MDB channelling and real L&D money is deep, not just a matter of scale.

Private and charity money is allowed under the Governing Instrument's "wide variety of sources" (UNFCCC, 2024). These cannot fill the gap. Global private charity for climate adaptation totals only a few tens of millions yearly.

Outside the current rules, the four-year World Bank host review is the main chance to rethink the design. A new Governing Instrument could add a clause for solidarity levies, a review trigger for revenue levels, or a route to an independent host. These require a COP decision, not a change to the Paris Agreement text (Schalatek, 2025). The Country-Coalition Matrix shows that rich nations moved from firm opposition to soft opposition: they accept L&D in theory but

strip it of its financial power. Whether running the Fund, plus rising physical losses, will push them towards mandatory payments is the key political question for the next talks. This study does not answer that. It clearly frames it: the task is to change the legal text limiting fiscal power, not to boost the Fund's standing.

Situating the Findings in the Literature

The findings confirm Calliari's (2018) discourse analysis of party positions in pre-Paris negotiations. The liability/cooperation divide she documented in that period has been reproduced in the FRLD's institutional architecture exactly as her analysis predicted. This study extends her account longitudinally: the same divide operates across the full period from 1991 to 2024, and it produces measurable fiscal consequences at the level of the Governing Instrument.

The findings confirm Gewirtzman et al.'s (2018) argument that conventional instruments are inadequate for L&D. This study extends that argument by showing that the instruments that would be adequate have been systematically rejected over 33 years, not merely overlooked. The rejected instruments register provides the first comprehensive record of this pattern across the full history of negotiations.

On institutional design, this study fills the gap identified in the existing literature, which characterised the liability exclusion in Decision 1/CP.21 as politically significant but did not trace its translation into the specific clauses of the FRLD Governing Instrument. The clause-level fiscal authority assessment does this directly, establishing a documented chain from the 2015 negotiating outcome to the 2023 institutional design.

CONCLUSION

This thesis tracked the growth of Loss and Damage governance under the UNFCCC from 1991 to 2024. Four methods were used: keyword counting, a Country-Coalition Position Matrix, an Instrument Evolution Table, and a fiscal authority assessment of the FRLD Governing Instrument.

The original 1991 AOSIS proposal set liability and mandatory payments. Subsequent phases steadily moved away from firm obligation. The 2010 Cancún Framework folded L&D into adaptation. The 2013 WIM limited the body to learning and discussion. The 2023 FRLD Governing Instrument plainly ruled out liability in its founding text (UNFCCC, 2024). Analysis of all texts shows a sharp drop in liability-linked words ("compensation," "restitution," "polluter pays," "state responsibility") and a clear rise in cooperation terms since 1991.

The Country-Coalition Matrix shows changing demands among developing country blocs. AOSIS and the LDC Group shifted from demanding mandatory, emissions-based pay-outs in 1991 to accepting a body built purely on voluntary gifts in 2023 (Falzon et al., 2023). This did not signal a change in their moral view. Their papers kept citing historical responsibility and climate justice (Calliari, 2018). It showed a practical choice: a formal body with weak money power was better than nothing. Developed country blocs moved from strong opposition, rejecting L&D as a separate area, to mild opposition: accepting the body while cutting its financial power at every meeting.

The Instrument Evolution Table lists six main finance ideas from 1991 to 2023 that were stopped or weakened. Each had mandatory parts, and each was blocked due to liability or donor resistance. From the 1991 mandatory payment pool to the 2013 dedicated financial body, and then to the 2021 Glasgow Loss and Damage Facility, financial ambition formally dropped at each COP. Needs did not lessen. Political room for mandatory tools shrank.

The close assessment of the FRLD Governing Instrument finds no clause giving the Fund power to set mandatory funding levels, impose fees, or link gifts to past emissions. The liability exclusion in Decision 1/CP.21 closely mirrors the absence of money power in the Fund's founding text. The Governing Instrument sets no funding floor, defines no mandatory payment rule, and provides no way to penalise non-paying parties (UNFCCC, 2024).

The answer to the main research question is simple: the norm shift in UNFCCC texts from liability to cooperation between 1991 and 2024 strongly limits the FRLD's power to enforce mandatory revenue measures. Proof from all four methods supports this. The FRLD's funding gap is not a brief political issue waiting for new governments or better pushing. It is a fixed feature of a body built on a legal structure that plainly rules out the basis for mandatory giving (Wewerinke-Singh & Salili, 2020).

This work offers three results.

The first point is for the climate governance study. This is the first long-term, factual report on the liability-to-cooperation norm shift in L&D governance, covering the entire 1991 to 2024 period, with clear keyword data and a full list of rejected tools. Calliari (2018) and Vanhala and Hestbaek (2016) described the shift in general and at certain meeting times. This study provides quantitative data on its path and turning points across the entire history of the regime, moving the field from general writing to proven evidence.

The second point is for institutional design theory. The idea of destructive ambiguity clarifies the usual notion of constructive ambiguity in global talks. Constructive ambiguity is usually seen as helping gradual norm growth. This study finds that when logic turns around, when the argument is about money-raising power, unclear language always means the minimum, and the minimum is voluntary cooperation. Liability needs a positive legal text to work. Just the lack of such text is enough to make voluntary cooperation rule. This idea can apply to other global fund designs where money power is disputed.

The third point is for public finance and institutional economics. The close money power check of the FRLD Governing Instrument shows that the Fund's design prevents it from raising the amount of money it was told to provide. Money power, not political will or technical skill, is vital for securing sufficient funding from global grant-based funds. A well-run, skilled fund working on voluntary pledges will still yield too little and too unstable income if its founding document gives it no power to require gifts. Enough money for L&D cannot be reached in the current FRLD design without adding a mandatory duty that does not trigger the Paris Agreement's liability limits. This is the main outcome of this study.

The four-year World Bank hosting review is thus not a routine management task. It is the first formal chance to ask plainly if the Governing Instrument needs change. Two questions must be asked. First: Should the Governing Instrument include a clause allowing solidarity fees on

international air or sea transport? These fees charge economic action rather than assigning past blame, meaning they do not break paragraph 51 of Decision 1/CP.21. Second: should the Governing Instrument add a funding-adequacy review trigger, needing the Board to tell the COP when actual funding falls below a set level compared to judged need? Without one of these tools, the Fund will stay a sign of political recognition for the L&D issue but lack the financial strength to act.

The second outcome concerns the negotiating method. Developing countries accepted a Governing Instrument without enough money or power, partly because of the changing demand pattern this study shows, as each past loss of money or power narrowed the room for the next round. Weak text is hard to reopen (Calliari, 2018). Advocates entering the World Bank review, the next funding talks, or any future COP chat on new finance should treat the money power question as the main goal. Accepting vague terms for "solidarity contributions" or "innovative sources" without a clause that spells out how such sources can be used risks, with talk processes and technical work replacing capital commitments repeatedly.

The FRLD started working at the end of 2023. Meaningful pay-out data will not be ready until at least 2027. The structural argument here rests on a pre-start institutional design check rather than a post-start operational outcome. A few large donors make big voluntary gifts that partly make up for the lack of legal duty. If that happens, it would challenge the structural argument and suggest that political will can replace legal authority in specific cases. A follow-up study, once payout data is ready, should test this directly and either confirm or alter the structural claim.

The four-year World Bank-hosting review should be studied as a governance negotiation in its own right. Using the Country-Coalition Matrix method on the review would test if the norm and position patterns shown here repeat in a more focused setting, or if running the Fund changes any developed country positions. If the EU and the Umbrella Group keep mild opposition during the review, the thesis's structural point is further confirmed.

Lastly, this study focuses on the formal UNFCCC system and does not show the full picture of L&D-relevant money. Bilateral deals, South-South assistance, and MDB programs may provide support outside the FRLD, thereby reducing the real funding gap, even if the formal structure remains weak. A study mapping these parallel flows, their size, their terms, and their recipients, would give a fuller account of what money truly reaches climate-hit communities and how far the total falls from what is needed.

REFERENCES

- Åberg, A., & Jeffs, N. (2022). Loss and Damage finance in the climate negotiations: Key challenges and next steps. Royal Institute of International Affairs.
<https://doi.org/10.55317/9781784135461>
- Adeniyi, D. (2025, January 29). The broken promise of loss and damage finance: Can Europe help? ECDPM. <https://ecdpm.org/work/broken-promise-loss-and-damage-finance-can-europe-help>
- Adger, W. (2023). Loss and Damage from climate change: Legacies from Glasgow and Sharm el-Sheikh. *Scottish Geographical Journal*, 139, 142–149.
<https://doi.org/10.1080/14702541.2023.2194285>
- AIIB, European Bank, & Paulson Institute. (2025). Unlocking Private Finance for Nature as Infrastructure. https://www.aiib.org/en/news-events/news/2025/_pdf/Unlocking-Private-Finance-for-Nature-as-Infrastructure11.15.pdf
- Aleksandrova, M., & Costella, C. (2021). Reaching the poorest and most vulnerable: Addressing loss and damage through social protection. *CURRENT OPINION IN ENVIRONMENTAL SUSTAINABILITY*, 50, 121–128. (WOS:000696958300013).
<https://doi.org/10.1016/j.cosust.2021.03.010>
- Allen, S., Mackey, A., & Witton, R. (2024, March 5). Leave No Mountain Behind: The Synthesis Series—Integration of Indigenous knowledge for adaptation in mountain regions—weADAPT. <https://weadapt.org/knowledge-base/adaptation-in-mountains/leave-no-mountain-behind-the-synthesis-series-integration-of-indigenous-knowledge-for-adaptation-in-mountain-regions/>
- AOSIS. (2012). Submission by the Republic of Nauru on Behalf of the Alliance of Small Island States (AOSIS). Plan of Work for the Ad-Hoc Working Group on the Durban Platform for Enhanced Action. <http://aosis.org/wp-content/uploads/2013/05/AOSIS-Submission-ADP-Final-May-2012.pdf>
- Bahinipati, C. (2020). Assessing the costs of droughts in rural India: A comparison of economic and non-economic loss and damage. *CURRENT SCIENCE*, 118(11), 1832–1841. (WOS:000538170500033). <https://doi.org/10.18520/cs/v118/i11/1832-1841>

Bahinipati, C., Patnaik, U., & Senapati, A. (2024). Building resilience to floods in India: Human development, income, inequality and forest cover. *CURRENT SCIENCE*, 127(8), 931–939. (WOS:001343402300017). <https://doi.org/10.18520/cs/v127/i8/931-939>

Barrett, S. (2003). *Environment and statecraft: The strategy of environmental treaty-making*. Oxford University Press.

Bhandari, P., Madan, N., Prasad, R., Bhandari, R., Sharma, V., Dey, S., Mandal, K. K., Scott, C., & Ramanathan, G. (2022). Climate Resilience Landscape for India: How philanthropies can fill the financing gap.

Bhowmik, J., Irfanullah, H., & Selim, S. (2021). Empirical evidence from Bangladesh of assessing climate hazard-related loss and damage and state of adaptive capacity to address them. *CLIMATE RISK MANAGEMENT*, 31. (WOS:000621750100002). <https://doi.org/10.1016/j.crm.2021.100273>

Board of the FRLD. (2024a). Co-Chairs’ proposal on the operationalization of the Fund as a World Bank-hosted financial intermediary fund (Board Document FLD/B.3/7). Fund for responding to Loss and Damage. <https://unfccc.int/event/third-meeting-of-the-board-of-the-fund-for-responding-to-loss-and-damage>

Board of the FRLD. (2024b). Decisions of the Board – Third meeting of the Board, 18–20 September 2024 (Decision Document FRLD/B.3/11). Fund for Responding to Loss and Damage. <https://unfccc.int/event/third-meeting-of-the-board-of-the-fund-for-responding-to-loss-and-damage>

Board of the FRLD. (2024c). Report of the third meeting of the Board of the Fund for responding to Loss and Damage (Meeting Report FRLD/B.3/12). Fund for responding to Loss and Damage. https://unfccc.int/sites/default/files/resource/B3_Board_meeting%20report.pdf

Board of the FRLD. (2024d). Report on the launch event of the high-level dialogue (Board Document FRLD/B.4/12). Fund for responding to Loss and Damage.

Board of the FRLD. (2025a). Decisions of the Board – seventh meeting of the Board, 7–9 October 2025 (Decision Document FRLD/B.7/11). Fund for responding to Loss and Damage.

Board of the FRLD. (2025b). Guidelines of Participation of Advisers in Board Meetings and Meetings of Committees (Document FRLD/B.5/9). Fund for Responding to Loss and Damage.

Board of the FRLD. (2025c). Proposal for the start-up phase of the Fund for responding to Loss and Damage (Board Document FRLD/B.5/7). Fund for responding to Loss and Damage (FRLD). <https://unfccc.int/>

Board of the FRLD. (2025d). Report of the fifth meeting of the Board of the Fund for responding to Loss and Damage (Meeting Report FRLD/B.5/14). Fund for responding to Loss and Damage.

Board of the FRLD. (2026a). Proposal for a country support system (Board Document FRLD/B.8/12). Fund for responding to Loss and Damage.

Board of the FRLD. (2026b). Report of the seventh meeting of the Board of the Fund for responding to Loss and Damage (Meeting Report FRLD/B.7/12). Fund for responding to Loss and Damage.

Board of the FRLD. (2026c). Report on the activities of the Secretariat (Board Document FRLD/B.8/3). Fund for responding to Loss and Damage. <https://www.frlld.org/>

Board of the FRLD & IBRD. (2024, October). Agreement on the Hosting Arrangements for the Secretariat of the Fund for Responding to Loss and Damage between the Board of the Fund for Responding to Loss and Damage and International Bank for Reconstruction and Development. World Bank. <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099102524142518821>

Boda, C., Faran, T., Scown, M., Dorkenoo, K., Chaffin, B., Nastar, M., & Boyd, E. (2021). Loss and damage from climate change and implicit assumptions of sustainable development. *CLIMATIC CHANGE*, 164(1–2). (WOS:000612855100002). <https://doi.org/10.1007/s10584-021-02970-z>

Boyd, E., Chaffin, B., Dorkenoo, K., Jackson, G., Harrington, L., N’Guetta, A., Johansson, E., Nordlander, L., De Rosa, S., Raju, E., Scown, M., Soo, J., & Stuart-Smith, R. (2021). Loss and damage from climate change: A new climate justice agenda. *ONE EARTH*, 4(10), 1365–1370. (WOS:000711140400008). <https://doi.org/10.1016/j.oneear.2021.09.015>

C3S. (2025). 2024 is the first year to exceed 1.5°C above pre-industrial level. Copernicus Climate Change Service. <https://climate.copernicus.eu/copernicus-2024-first-year-exceed-15degc-above-pre-industrial-level>

Calliari, E. (2014). Loss & Damage: A Critical Discourse Analysis. <https://doi.org/10.2139/SSRN.2510785>

Calliari, E. (2018a). Loss and damage: A critical discourse analysis of Parties' positions in climate change negotiations. *JOURNAL OF RISK RESEARCH*, 21(6), 725–747. (WOS:000433987300004). <https://doi.org/10.1080/13669877.2016.1240706>

Calliari, E. (2018b). Loss and damage: A critical discourse analysis of Parties' positions in climate change negotiations. *Journal of Risk Research*, 21, 725–747. <https://doi.org/10.1080/13669877.2016.1240706>

Calliari, E., Serdeczny, O., & Vanhala, L. (2020). Making sense of the politics in the climate change loss & damage debate. *GLOBAL ENVIRONMENTAL CHANGE-HUMAN AND POLICY DIMENSIONS*, 64. (WOS:000577629200009). <https://doi.org/10.1016/j.gloenvcha.2020.102133>

Calliari, E., Surminski, S., & J. Mysiak. (2018). The Politics of (and Behind) the UNFCCC's Loss and Damage Mechanism. *Loss and Damage from Climate Change*. https://doi.org/10.1007/978-3-319-72026-5_6

Calvin, K., Dasgupta, D., Krinner, G., Mukherji, A., Thorne, P. W., Trisos, C., Romero, J., Aldunce, P., Barrett, K., Blanco, G., Cheung, W. W. L., Connors, S., Denton, F., Diongue-Niang, A., Dodman, D., Garschagen, M., Geden, O., Hayward, B., Jones, C., ... Péan, C. (with Lee, H.). (2023). IPCC, 2023: Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds.)]. IPCC, Geneva, Switzerland. (First). Intergovernmental Panel on Climate Change (IPCC). <https://doi.org/10.59327/IPCC/AR6-9789291691647>

Case Concerning the Factory at Chorzów (Germany v. Poland), Series A, No. 17 P.C.I.J. 4 (1928).

Clarke, B., Otto, F., & Jones, R. (2023). When don't we need a new extreme event attribution study? *CLIMATIC CHANGE*, 176(5). (WOS:000983252200001).
<https://doi.org/10.1007/s10584-023-03521-4>

CPI. (2023). Global Landscape of Climate Finance 2023. Climate Policy Initiative.
<https://www.climatepolicyinitiative.org/wp-content/uploads/2023/11/Global-Landscape-of-Climate-Finance-2023.pdf>

D. Coumou, Paola A. Arias, A. Bastos, Charlotte Kendra Gotangco Gonzales, G. C. Hegerl, Pandora Hope, Chris Jack, Friederike E. L. Otto, Fahad Saeed, O. Serdeczny, Theodore G. Shepherd, & R. Vautard. (2024). How can event attribution science underpin financial decisions on Loss and Damage? *PNAS Nexus*, 3. <https://doi.org/10.1093/pnasnexus/pgae277>

Doktycz, C., & Abkowitz, M. (2019). Loss and Damage Estimation for Extreme Weather Events: State of the Practice. *SUSTAINABILITY*, 11(15). (WOS:000485230200248).
<https://doi.org/10.3390/su11154243>

Dorkenoo, K., Scown, M., & Boyd, E. (2022). A critical review of disproportionality in loss and damage from climate change. *WILEY INTERDISCIPLINARY REVIEWS-CLIMATE CHANGE*, 13(4). (WOS:000759768400001). <https://doi.org/10.1002/wcc.770>

Fairclough, N. (1995). *Critical Discourse Analysis: The Critical Study of Language*. Longman. <https://www.felsemiotica.com/descargas/Fairclough-Norman-Critical-Discourse-Analysis.-The-Critical-Study-of-Language.pdf>

Fairclough, N. (2003). *Analysing Discourse: Textual Analysis for Social Research*. Routledge. <https://howardaudio.wordpress.com/wp-content/uploads/2018/01/n-fairclough-analysing-discourse.pdf>

Falzon, D., Shaia, F., Roberts, J., Hossain, M., Robinson, S., Khan, M., & Ciplet, D. (2023). Tactical Opposition: Obstructing Loss and Damage Finance in the United Nations Climate Negotiations (vol 23, pg 95, 2023). *GLOBAL ENVIRONMENTAL POLITICS*, 23(4), 222–222. (WOS:001106852200007). https://doi.org/10.1162/glep_x_00741

Fekete, A., & Sakdapolrak, P. (2014). Loss and Damage as an Alternative to Resilience and Vulnerability? Preliminary Reflections on an Emerging Climate Change Adaptation Discourse. *International Journal of Disaster Risk Science*, 5(1), 88–93. (WOS:000353357400009). <https://doi.org/10.1007/s13753-014-0012-7>

Fezzigna, P., Wens, M., Gupta, J., & Scussolini, P. (2025). Data injustice and attribution of drought events: Implications for global climate policy. *CLIMATE POLICY*. (WOS:001417930200001). <https://doi.org/10.1080/14693062.2025.2462633>

Fischer, E. M., & Knutti, R. (2015). Anthropogenic contribution to global occurrence of heavy-precipitation and high-temperature extremes. *Nature Climate Change*, 5(6), 560–564. <https://doi.org/10.1038/nclimate2617>

Fritsch, W., Kraemer, M., & Songwe, V. (2025). Beyond Private Finance: The Limits of Capital Markets in Financing EMDE Transition. <https://unfccc.int/sites/default/files/resource/Beyond%20Private%20Finance%20%282025%29%20Fritsch%2C%20Kraemer%2C%20and%20Songwe%20V.2.pdf>

G77 and China. (2013a). Approaches to Address Loss and Damage Associated with Climate Change Impacts in Developing Countries That Are Particularly Vulnerable to the Adverse Effects of Climate Change to Enhance Adaptive Capacity. Submission from the Group of 77 and China. <http://unfccc.int/resource/docs/2013/sbi/eng/crp01.pdf>

G77 and China. (2013b). Input from the G77 & China in Preparation for COP19 on Loss and Damage Associated with Climate Change Impacts in Developing Countries That Are Particularly Vulnerable to the Adverse Effects of Climate Change. https://unfccc.int/files/adaptation/application/pdf/ld_g77_submission_nov_2013.pdf

García-Portela, L. (2023). Backward-Looking Principles of Climate Justice: The Unjustified Move from the Polluter Pays Principle to the Beneficiary Pays Principle. *Res Publica* (Liverpool, England), 29(3), 367–384. <https://doi.org/10.1007/s11158-022-09569-w>

García-Portela, L. (2024). A Minimal Capabilities-Based Account of Loss and Damage. *JOURNAL OF HUMAN DEVELOPMENT AND CAPABILITIES*, 25(1), 170–193. (WOS:001145764100001). <https://doi.org/10.1080/19452829.2024.2305388>

Germanwatch e.V. (2025). Climate Risk Index 2025: Who suffers most from extreme weather events? <https://www.germanwatch.org/sites/default/files/2025-02/Climate%20Risk%20Index%202025.pdf>

Gewirtzman, J., Natson, S., Richards, J., Hoffmeister, V., Durand, A., Weikmans, R., Huq, S., & Roberts, J. (2018). Financing loss and damage: Reviewing options under the Warsaw

International Mechanism. CLIMATE POLICY, 18(8), 1076–1086.

(WOS:000443922200010). <https://doi.org/10.1080/14693062.2018.1450724>

Hegglin, M., Bastos, A., Bovensmann, H., Buchwitz, M., Fawcett, D., Ghent, D., Kulk, G., Sathyendranath, S., Shepherd, T., Quegan, S., Röthlisberger, R., Briggs, S., Buontempo, C., Cazenave, A., Chuvieco, E., Ciais, P., Crisp, D., Engelen, R., Fadnavis, S., ... Woolway, R. (2022). Space-based Earth observation in support of the UNFCCC Paris Agreement.

FRONTIERS IN ENVIRONMENTAL SCIENCE, 10. (WOS:000875618500001).

<https://doi.org/10.3389/fenvs.2022.941490>

Heinrichs, P., Brandimarte, I., & Lerner, A. B. (2025). Climate (In)justice and the Residual Category of Non-Economic Loss and Damage. *Millennium*, 53, 843–867.

<https://doi.org/10.1177/03058298251366196>

Huber, J., & Murray, U. (2024). Turning climate justice into practice? Channeling loss and damage funding through national social protection systems in climate-vulnerable countries. *WILEY INTERDISCIPLINARY REVIEWS-CLIMATE CHANGE*, 15(2).

(WOS:001099449800001). <https://doi.org/10.1002/wcc.867>

Ike Uri, Stacy-ann Robinson, J. T. Roberts, D. Ciptet, R. Weikmans, & Mizan R. Khan. (2024). Equity and Justice in Loss and Damage Finance: A Narrative Review of Catalysts and Obstacles. *Current Climate Change Reports*, 10, 33–45. <https://doi.org/10.1007/s40641-024-00196-6>

Intergovernmental Negotiating Committee. (1991). Vanuatu: Draft Annex Relating to Article 23 (Insurance) for Inclusion in the Revised Single Text on Elements Relating to Mechanisms. INC for a Framework Convention on Climate Change.

<https://unfccc.int/sites/default/files/resource/docs/a/wg2crp08.pdf>

Klein, R. J. T., Midgley, G. F., Preston, B. L., Alam, M., Berkhout, F. G. H., Dow, K., & Shaw, M. R. (2014). Adaptation Opportunities, Constraints, and Limits. In C. B. Field, V. R. Barros, D. J. Dokken, K. J. Mach, M. D. Mastrandrea, T. E. Bilir, M. Chatterjee, K. L. Ebi, Y. O. Estrada, R. C. Genova, B. Girma, E. S. Kissel, A. N. Levy, S. MacCracken, P. R. Mastrandrea, & L. L. White (Eds.), *Climate Change 2014: Impacts, Adaptation, and Vulnerability. Part A: Global and Sectoral Aspects. Contribution of Working Group II to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* (pp. 899–943). Cambridge University Press.

- Kumar, V. (2024). Financing loss and damage and international climate governance: A climate justice perspective. *International Journal of Political Science and Governance*. <https://doi.org/10.33545/26646021.2024.v6.i2c.379>
- Limsakul, A., Kammuang, A., Paengkaew, W., Sooktawee, S., & Aroonchan, N. (2024). Changes in slow-onset climate events in Thailand. *ENVIRONMENTAL ENGINEERING RESEARCH*, 29(1). (WOS:001080440400012). <https://doi.org/10.4491/eer.2022.784>
- Markandya, A., & González-Eguino, M. (2019). Integrated Assessment for Identifying Climate Finance Needs for Loss and Damage: A Critical Review: Concepts, Methods and Policy Options (pp. 343–362). https://doi.org/10.1007/978-3-319-72026-5_14
- Martyr-Koller, R., Thomas, A., Schleussner, C., Nauels, A., & Lissner, T. (2021). Loss and damage implications of sea-level rise on Small Island Developing States. *CURRENT OPINION IN ENVIRONMENTAL SUSTAINABILITY*, 50, 245–259. (WOS:000696958300026). <https://doi.org/10.1016/j.cosust.2021.05.001>
- Mcdonnell, S. (2023). The COP27 decision and future directions for loss and damage finance: Addressing vulnerability and non-economic loss and damage. *REVIEW OF EUROPEAN COMPARATIVE & INTERNATIONAL ENVIRONMENTAL LAW*, 32(3), 416–427. (WOS:001101727400001). <https://doi.org/10.1111/reel.12521>
- McGrath, M. (2013, November 11). Typhoon prompts “fast” by Philippines climate delegate. *BBC News*. <https://www.bbc.com/news/science-environment-24899647>
- Mechler, R., Bouwer, L. M., Schinko, T., Surminski, S., & Linnerooth-Bayer, J. (Eds.). (2019). *Loss and Damage from Climate Change: Concepts, Methods and Policy Options*. Springer International Publishing. <https://doi.org/10.1007/978-3-319-72026-5>
- Mechler, R., & Deubelli, T. (2021). Finance for Loss and Damage: A comprehensive risk analytical approach. *CURRENT OPINION IN ENVIRONMENTAL SUSTAINABILITY*, 50, 185–196. (WOS:000696958300020). <https://doi.org/10.1016/j.cosust.2021.03.012>
- Mechler, R., Singh, C., Ebi, K., Djalante, R., Thomas, A., James, R., Tschakert, P., Wewerinke-Singh, M., Schinko, T., Ley, D., Nalau, J., Bouwer, L., Huggel, C., Huq, S., Linnerooth-Bayer, J., Surminski, S., Pinho, P., Jones, R., Boyd, E., & Revi, A. (2020). *Loss and Damage and limits to adaptation: Recent IPCC insights and implications for climate*

science and policy. SUSTAINABILITY SCIENCE, 15(4), 1245–1251.
(WOS:000530229800001). <https://doi.org/10.1007/s11625-020-00807-9>

MunichRE. (2025, January). Natural Disasters 2024 | Factsheet.
https://www.munichre.com/content/dam/munichre/mrwebsitespressreleases/MunichRe-NatCAT-Stats2024-Full-Year-Factsheet.pdf/_jcr_content/renditions/original./MunichRe-NatCAT-Stats2024-Full-Year-Factsheet.pdf

Nand, M., Clissold, R., McNamara, K., & Yee, M. (2024). Guiding our responses to climate change by what people value: Insights from Fiji. GEOFORUM, 155.
(WOS:001286325100001). <https://doi.org/10.1016/j.geoforum.2024.104080>

Neslen, A., & Harvey, F. (2016, November 16). John Kerry: We will fight to keep US in the Paris climate deal. The Guardian.
<https://www.theguardian.com/environment/2016/nov/16/john-kerry-we-will-fight-to-keep-us-in-the-paris-climate-deal>

OECD. (2021). Managing Climate Risks, Facing up to Losses and Damages: Understanding, Reducing and Managing Risks. OECD Publishing. <https://doi.org/10.1787/55ea1cc9-en>

Ohdedar, B. (2016). Loss and Damage from the Impacts of Climate Change: A Framework for Implementation. Nordic Journal of International Law, 85, 1–36.
<https://doi.org/10.1163/15718107-08501001>

Pill, M. (2022). Towards a funding mechanism for loss and damage from climate change impacts. CLIMATE RISK MANAGEMENT, 35. (WOS:000776078700004).
<https://doi.org/10.1016/j.crm.2021.100391>

Richards, J.-A., & Schalatek, L. (2017). Financing Loss and Damage: A Look at Governance and Implementation Options.

Roberts, E., & Pelling, M. (2018). Climate change-related loss and damage: Translating the global policy agenda for national policy processes. CLIMATE AND DEVELOPMENT, 10(1), 4–17. (WOS:000428134000002). <https://doi.org/10.1080/17565529.2016.1184608>

Robinson, S., Khan, M., Roberts, J., Weikmans, R., & Ciplet, D. (2021). Financing loss and damage from slow onset events in developing countries. CURRENT OPINION IN ENVIRONMENTAL SUSTAINABILITY, 50, 138–148. (WOS:000696958300015).
<https://doi.org/10.1016/j.cosust.2021.03.014>

Santos, C. T., Toda, L., Orduña, J. R., Santos, F. D., & Ferrão, J. (2015). The impacts of Typhoon Haiyan in the Philippines: Implications to land use planning. *Climate, Disaster and Development Journal*, 1(1), 57–66. <https://doi.org/10.18783/cddj.v001.i01.a06>

Schalatek, L. (2025). Fund for Responding to Loss and Damage. In *Climate Finance Fundamentals* (13th ed., pp. 1–21). Heinrich Böll Foundation.

Schinko, T., Mechler, R., & Hochrainer-Stigler, S. (2018). The Risk and Policy Space for Loss and Damage: Integrating Notions of Distributive and Compensatory Justice with Comprehensive Climate Risk Management. *Loss and Damage from Climate Change*. https://doi.org/10.1007/978-3-319-72026-5_4

Scown, M., Du, H., Jackson, G., De Rosa, S., & Boyd, E. (2025). Towards CS4L&D: Advancing climate services for loss and damage. *CLIMATE SERVICES*, 38. (WOS:001470014200001). <https://doi.org/10.1016/j.cliser.2025.100563>

Siegele, L. (2023). Financing for loss and damage under the UNFCCC: Have we come full circle? *REVIEW OF EUROPEAN COMPARATIVE & INTERNATIONAL ENVIRONMENTAL LAW*, 32(3), 403–415. (WOS:001099012000001). <https://doi.org/10.1111/reel.12524>

Simlinger, F., & Mayer, B. (2018). Legal Responses to Climate Change Induced Loss and Damage. *Loss and Damage from Climate Change*. https://doi.org/10.1007/978-3-319-72026-5_7

Thomas, A., Baptiste, A., Martyr-Koller, R., Pringle, P., & Rhiney, K. (2020). Climate Change and Small Island Developing States. In A. Gadgil & T. Tomich (Eds.), *ANNUAL REVIEW OF ENVIRONMENT AND RESOURCES*, VOL 45 (Vol. 45, pp. 1–27). (WOS:000590409700001). <https://doi.org/10.1146/annurev-environ-012320-083355>

Trail Smelter Case (United States v. Canada), Vol. III Reports of International Arbitral Awards (RIAA) 1905 (Trail Smelter Arbitral Tribunal 1941). https://legal.un.org/riaa/cases/vol_III/1905-1982.pdf

UNDP. (2023, June 26). Loss and Damage | Climate Promise. <https://climatepromise.undp.org/what-we-do/areas-of-work/loss-and-damage>

UNEP. (2023a). Adaptation Gap Report 2023: Underfinanced. Underprepared. Inadequate investment and planning for climate adaptation leave the world exposed.

<https://doi.org/10.59117/20.500.11822/43796>

UNEP. (2023b, October 18). Adaptation | UNEP - UN Environment Programme.

<https://www.unep.org/topics/climate-action/adaptation>

UNEP. (2023c, October 18). Mitigation | UNEP - UN Environment Programme.

<https://www.unep.org/topics/climate-action/mitigation>

UNEP. (2023d, October 31). About Loss and damage | UNEP - UN Environment

Programme. <https://www.unep.org/topics/climate-action/loss-and-damage/about-loss-and-damage>

UNFCCC. (1991a). Consolidated working document. Addendum. Annex: V. Insurance mechanism. (A/AC.237/Misc.17/Add.9).

UNFCCC. (1991b). Negotiation of a Framework Convention on Climate Change. Elements relating to mechanisms. Vanuatu: Draft annex relating to article 23 (insurance) for inclusion in the revised single text on elements relating to mechanisms submitted by the Co-chairmen of Working Group II. (A/AC.237/WG.II/Misc.13).

<https://unfccc.int/resource/docs/a/wg2crp08.pdf>

UNFCCC. (1997). Kyoto Protocol to the UNFCCC.

https://treaties.un.org/doc/Treaties/1998/09/19980921%2004-41%20PM/Ch_XXVII_07_ap.pdf

UNFCCC. (2001). Decision 7/CP.7. Funding under the Convention.

(FCCC/CP/2001/13/Add.1).

https://unfccc.int/files/cooperation_and_support/financial_mechanism/application/pdf/7_cp.7.pdf

UNFCCC. (2010). Decision 1/CP.16, The Cancun Agreements: Outcome of the work of the Ad Hoc Working Group on Long-term Cooperative Action under the Convention (Issue FCCC/CP/2010/7/Add.1). UNFCCC Secretariat.

<https://unfccc.int/resource/docs/2010/cop16/eng/07a01.pdf>

UNFCCC. (2013). Decision 2/CP.19, Warsaw International Mechanism for Loss and Damage Associated with Climate Change Impacts (Issue FCCC/CP/2013/10/Add.1). UNFCCC Secretariat. <http://unfccc.int/resource/docs/2013/cop19/eng/10a01.pdf#page=6>

UNFCCC. (2014). Report of the Executive Committee of the Warsaw International Mechanism for Loss and Damage associated with Climate Change Impacts (FCCC/SB/2014/4). UNFCCC.

UNFCCC. (2015a). Synthesis report on the aggregate effect of the intended nationally determined contributions (FCCC/CP/2015/7).

UNFCCC. (2015b, December). Paris Agreement. https://unfccc.int/sites/default/files/english_paris_agreement.pdf

UNFCCC. (2016a). Decision 1/CP.21, Adoption of the Paris Agreement (Issue FCCC/CP/2015/10/Add.1). UNFCCC Secretariat. <https://unfccc.int/resource/docs/2015/cop21/eng/10a01.pdf>

UNFCCC. (2016b). Report of the Conference of the Parties on its twenty-first session, held in Paris from 30 November to 13 December 2015 (FCCC/CP/2015/10/Add.1). UNFCCC.

UNFCCC. (2017). Report of the Executive Committee of the Warsaw International Mechanism for loss and damage associated with climate change impacts. Addendum (Issue FCCC/SB/2017/1/Add.1). UNFCCC Secretariat. <http://unfccc.int/resource/docs/2017/sb/eng/105.pdf>

UNFCCC. (2019). Decision -/CMA.4. Santiago network for averting, minimizing and addressing loss and damage under the Warsaw International Mechanism for Loss and Damage associated with Climate Change Impacts. https://unfccc.int/sites/default/files/resource/cma4_auv_7_WIM.pdf

UNFCCC. (2023). Report of the Conference of the Parties on Its Twenty-Seventh Session, Held in Sharm el-Sheikh from 6 to 20 November 2022 (Issue FCCC/CP/2022/10). UNFCCC Secretariat.

UNFCCC. (2024). Report of the Conference of the Parties on its twenty-eighth session, held in the United Arab Emirates from 30 November to 13 December 2023—Addendum Part two: Action taken by the Conference of the Parties at its twenty-eighth session (FCCC/CP/2023/11/Add.1).

UNFCCC. (2025a). Report of the Conference of the Parties on its twenty-ninth session, held in Baku from 11 to 24 November 2024 (FCCC/CP/2024/11/Add.2; Issue FCCC/CP/2022/10). UNFCCC Secretariat.

UNFCCC. (2025b, November). Report on the Baku to Belem Roadmap: Roadmap to 1.3T. https://unfccc.int/sites/default/files/resource/Relatorio_Roadmap_COP29_COP30_EN_final.pdf

UNFCCC Secretariat. (2012). A literature review on a range of approaches to address loss and damage associated with the adverse effects of climate change (Issue FCCC/SBI/2012/INF.14). UNFCCC Secretariat. <http://unfccc.int/resource/docs/2012/sbi/eng/inf14.pdf>

United Nations. (1992). United Nations Framework Convention on Climate Change. United Nations. <https://unfccc.int/sites/default/files/conveng.pdf>

van Schie, D., Jackson, G., Ranon, R., Mirza, A., Hossain, M., Bakthaoui, I., & Anderson, S. (2024). Addressing non-economic loss and damage: Learning from autonomous responses in Bangladesh. *CLIMATIC CHANGE*, 177(8). (WOS:001274057200001). <https://doi.org/10.1007/s10584-024-03782-7>

Vanhala, L. (2023). Putting the constructive ambiguity of climate change loss and damage into practice: The early work of the UNFCCC WIM ExCom. *REVIEW OF EUROPEAN COMPARATIVE & INTERNATIONAL ENVIRONMENTAL LAW*, 32(3), 428–438. (WOS:001017526700001). <https://doi.org/10.1111/reel.12508>

Vanhala, L., & Hestbaek, C. (2016). Framing Climate Change Loss and Damage in UNFCCC Negotiations. *Global Environmental Politics*, 16(4), 111–129. https://doi.org/10.1162/GLEP_a_00379

Verchick, R. (2018). Can “loss and damage” carry the load? *PHILOSOPHICAL TRANSACTIONS OF THE ROYAL SOCIETY A-MATHEMATICAL PHYSICAL AND ENGINEERING SCIENCES*, 376(2119). (WOS:000429046300017). <https://doi.org/10.1098/rsta.2017.0070>

Vidal, J. (2015, December 16). How a “typo” nearly derailed the Paris climate deal. *The Guardian*. <https://www.theguardian.com/environment/blog/2015/dec/16/how-a-typo-nearly-derailed-the-paris-climate-deal>

Wallimann-Helmer, I. (2023). Resilience and Nonideal Justice in Climate Loss and Damage Governance. *GLOBAL ENVIRONMENTAL POLITICS*, 23(3), 52–70. (WOS:001062360000004). https://doi.org/10.1162/glep_a_00723

Wallimann-Helmer, I., Meyer, L. H., Mintz-Woo, K., T. Schinko, & O. Serdeczny. (2018). The Ethical Challenges in the Context of Climate Loss and Damage. *Loss and Damage from Climate Change*. https://doi.org/10.1007/978-3-319-72026-5_2

Warner, K., & van der Geest, K. (2013). Loss and damage from climate change: Local-level evidence from nine vulnerable countries. *INTERNATIONAL JOURNAL OF GLOBAL WARMING*, 5(4), 367–386. (WOS:000328550200001). <https://doi.org/10.1504/IJGW.2013.057289>

Wewerinke-Singh, M. (2023). The Rising Tide of Rights: Addressing Climate Loss and Damage through Rights-Based Litigation. *TRANSNATIONAL ENVIRONMENTAL LAW*, 12(3), 537–566. (WOS:001172032900004). <https://doi.org/10.1017/S2047102523000183>

Wewerinke-Singh, M., & Salili, D. (2020). Between negotiations and litigation: Vanuatu's perspective on loss and damage from climate change. *CLIMATE POLICY*, 20(6), 681–692. (WOS:000547065500003). <https://doi.org/10.1080/14693062.2019.1623166>

WMO. (2024, November 8). 2024 is on track to be hottest year on record as warming temporarily hits 1.5°C. <https://wmo.int/news/media-centre/2024-track-be-hottest-year-record-warming-temporarily-hits-15degc>

World Bank. (2025). FRLD Trustee Report—12-31-2025: Financial Status of the Fund for Responding to Loss and Damage Trust Fund [Trustee Report]. International Bank for Reconstruction and Development (IBRD). <https://fiftrustee.worldbank.org/content/dam/fif/funds/frld/TrusteeReports/FRLD%20Trustee%20Report%20-%2012-31-2025.pdf>

Yvonne Karimi-Schmidt. (2020). The Issues of Loss and Damage Within the International Climate Law. *Unknown Journal*. <https://doi.org/10.2139/ssrn.3732980>

ANNEXURES

Table of all Financial Instruments Proposed

Year	Proposing Party	Instrument Type / Action	Status / Outcome	Description / Grounds for Dilution
1991	AOSIS / Vanuatu	International Insurance Pool	<i>Rejected / Proposed</i>	Based on mandatory contributions (GNP/\$CO ₂ 2\$); dropped from UNFCCC 1992 for liability concerns.
2001	AOSIS / G77	Compensation & Rehabilitation Fund	<i>Rejected</i>	Subsumed into SCCF as an adaptation instrument; compensation element removed.
2008	AOSIS	Multi-Window Mechanism	<i>Rejected / Proposed</i>	Included insurance, rehab, and risk management. Led to WIM (2013) which lacked financial authority.
2012	Bolivia, China, et al.	Solidarity Fund	<i>Proposed</i>	Joint statement for a fund to compensate for permanent and irreversible damage.
2012-19	Tuvalu	National Legislative Acts	<i>Adopted</i>	Established "Te Kaniva" and Tuvalu Survival Fund for domestic L&D response.
2013	G77 and China	Dedicated Financial Facility	<i>Rejected</i>	Decision 2/CP.19 stripped financial components; "compensation" and "liability" expunged.
2013	UNFCCC Parties	Warsaw Int. Mechanism (WIM)	<i>Adopted</i>	Established to address L&D from extreme/slow-onset events via knowledge.
2014	LDC Group	Displacement Coordination Facility	<i>Under Negotiation</i>	Proposal to assist in organized migration and relocation.
2015	UNFCCC Parties	Paris Agreement (Article 8)	<i>Adopted</i>	Anchored L&D as a pillar, but explicitly excluded a basis for liability/compensation.
2016	Saul Luciano Lliuya	Lawsuit vs. RWE (Germany)	<i>Under Negotiation</i>	Seeking risk management costs in Peru proportional to historical emissions.
2017-21	Uruguay, Nepal, et al.	National NDC Mechanisms	<i>Proposed</i>	National pledges to develop L&D assessment and agricultural insurance.
2018	Vanuatu	Climate Damages Tax (CDT)	<i>Proposed</i>	Proposal for a tax on fossil fuel companies to shift costs to responsible actors.
2018	Marshall Islands	Tile Til Eo 2050 Strategy	<i>Adopted</i>	Strategy identifying asset elevation and relocation as national responses.
2018-21	Bangladesh	National L&D Mechanisms	<i>Adopted</i>	Efforts to establish adaptation practices (rice varieties, deep tube-wells).
2019	UNFCCC Parties	Santiago Network	<i>Adopted</i>	Created at COP25 to catalyse technical assistance for vulnerable nations.
2021	G77 and China	Glasgow Loss & Damage Facility	<i>Rejected</i>	Blocked by US/EU/Australia; replaced by the non-binding "Glasgow Dialogue."
2021-22	Scotland, Wallonia	Early Financial Pledges	<i>Pledged</i>	Scotland pledged £7 million as early support for L&D funding.
2022	COP / CMA	Funding Arrangements & Fund	<i>Adopted</i>	Decision 2/CP.27: Historic agreement to establish a dedicated L&D Fund.
2022	V20 and G7	Global Shield	<i>Adopted</i>	Launch of insurance-based solutions for extreme weather events.

2023	AOSIS / LDC Group	Independent FRLD	<i>Partially Rejected</i>	Pushed for mandatory targets; the outcome was voluntary/World Bank-hosted.
2023	COP / CMA	Operationalization of FRLD	<i>Adopted</i>	Decisions 1/CP.28 and 5/CMA.5: Approved Governing Instrument for the Fund.
2023	Transitional Committee	Recommendations Mandate	<i>Mandated</i>	Tasked with designing the structure and governance of the new Fund.
2023	UNDRR and UNOPS	Santiago Network Secretariat	<i>Adopted (MOU)</i>	MOU for hosting the secretariat and financial arrangements.
2023	Pacific Islands	Pacific Resilience Facility	<i>Established</i>	Regional initiative focused on responding to L&D in the Pacific.
2024	FRLD Board	Naming & 2024-25 Workplan	<i>Adopted</i>	Official name: 'Fund for responding to Loss and Damage' (FRLD).
2024	Philippines	Host Country for FRLD Board	<i>Adopted</i>	Selected as host, Republic Act 12019 granted the Fund legal personality.
2024	World Bank	Hosting & Trustee Agreement	<i>Adopted</i>	Formalised the World Bank as interim trustee (MTO No. 069040) for 4 years.
2024	Global Donors	Initial Capitalization	<i>Pledged</i>	Various nations offered financial contributions to establish the Fund.
2024	Adaptation Fund Board	Resource Mobilization	<i>Adopted</i>	USD 300M target and a global small grants aggregator program.
2025	FRLD Board	Barbados Modalities (BIM)	<i>Adopted</i>	Start-up phase: USD 250M for 2025-26 for grant-based local interventions.
2025	FRLD Board	2026 Workplan & Resource Strat	<i>Adopted</i>	Strategy for 2027 replenishment; access via AF, GEF, and GCF entities.
2025-26	Haiti, Jamaica, et al.	National Project Proposals	<i>Under Review</i>	Proposals for flood recovery/water resilience (up to USD 20M each).
2025-26	FRLD Secretariat	Organizational Structure	<i>Provisional</i>	Development of PACE, GBA, BFM units and assessment methodologies.

Table A.1: Table of all Financial Instruments Proposed